

6G · STUDY GUIDE · STATUS: OCTOBER 2026

6G and IMT-2030, explained

A visual guide for telecom engineers: IMT-2030, spectrum, air interface, AI, sensing, satellites and the road to 2030.

Telecom Knowledge Hub

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6G is not standardised yet. This guide describes the vision, the studies and the working agreements as of October 2026. Details will change until 3GPP Release 21 is complete.

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What 6G Is, and What It Is Not

The names, the players, the lessons from 5G and where things stand in October 2026.

1.1 Two names for one thing

ITU-R · IMT-2030

Sets the vision and the requirements

Evaluates the candidate technologies

Identifies spectrum at the WRC

Does not write the protocols

3GPP · 6G

Studies the technology: Release 20

Writes the specifications: Release 21

Radio, core and services

Submits the result to the ITU

IMT-2000 = 3G · IMT-Advanced = 4G · IMT-2020 = 5G · IMT-2030 = 6G

6G has two official names. The ITU calls it IMT-2030 and defines what it should be able to do. 3GPP writes the actual technical specifications and will introduce 6G in Release 21. A technology becomes 6G when 3GPP submits it to the ITU and the ITU accepts it as IMT-2030.

ITU-R M.2160 · 3GPP Release 20 and 21 planning

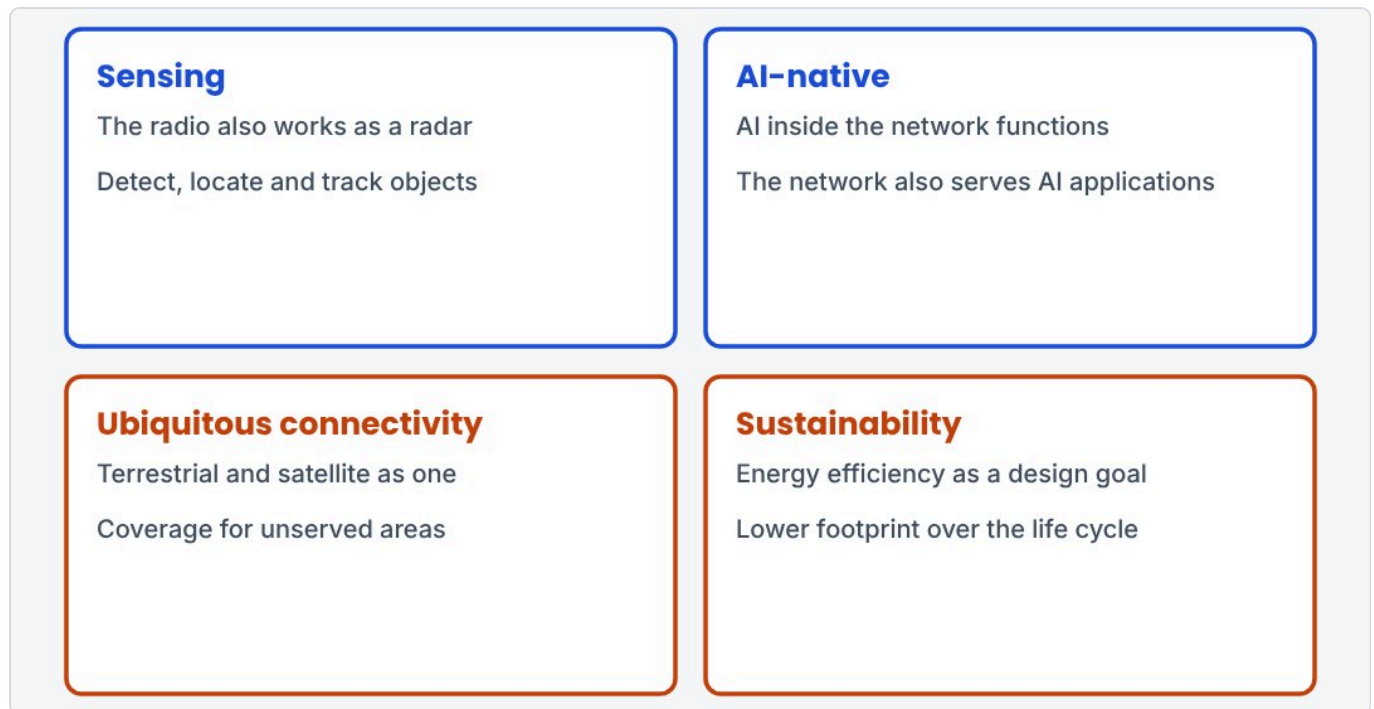
1.2 A generation every ten years

GEN	AROUND	RADIO	MAIN STEP
2G	1991	GSM	digital voice, SMS
3G	2001	UMTS / HSPA	mobile data
4G	2010	LTE / LTE-Advanced	mobile broadband, all IP
5G	2019	NR / 5G-Advanced	capacity, low latency, industry
6G	2030	specified in Release 21	sensing, AI, ubiquitous coverage

Mobile generations have arrived about every ten years. Each one is remembered for one main step. 6G is planned for commercial launch around 2030. Inside each generation the technology keeps evolving: 5G-Advanced, Releases 18 to 20, is the bridge to 6G.

3GPP release history · ITU-R IMT process

1.3 What is new compared with 5G



5G improved speed, latency and device density. 6G keeps improving those and adds capabilities that are new for a mobile network. The network should sense its surroundings, use AI as a built-in part, reach every place including through satellites, and do all that with less energy.

ITU-R M.2160 §4

1.4 What 6G is not

CLAIM	WHAT THE WORK SHOWS
"6G means terahertz"	first new bands are 7 to 15 GHz; sub-THz is research
"6G has a new waveform"	CP-OFDM stays, as in 5G
"6G is 100 times faster"	peak rates rise, but the goals are broader
"6G replaces 5G in 2030"	both coexist and share carriers
"6G exists today"	specifications are due at the end of 2028

Several popular claims do not match what is being standardised. 6G is not mainly a terahertz technology: the first focus is on bands between 7 and 15 GHz. It is not a completely new radio: it keeps OFDM. And it will not replace 5G overnight. Both will share spectrum for years.

3GPP Release 20 6G studies · WRC-27 agenda item 1.7

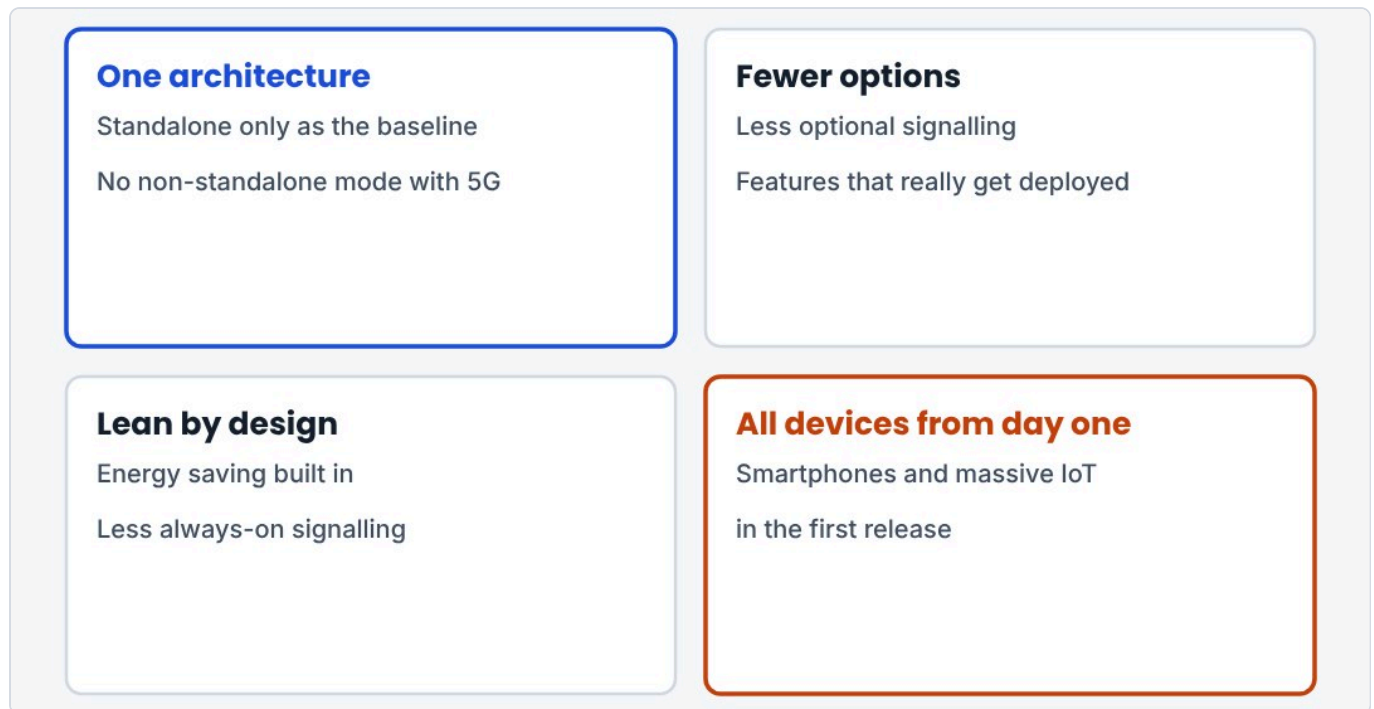
1.5 Who does what

BODY	ROLE
ITU-R Working Party 5D	IMT-2030 vision, requirements, evaluation
3GPP (RAN, SA, CT)	technical specifications
WRC-27	new frequency bands for IMT
ETSI, ATIS, ARIB, CCSA ...	3GPP partners, publish the specifications
6G-IA, Next G Alliance ...	regional research and positions
O-RAN ALLIANCE	open RAN interfaces, lower-layer split
GSMA, NGMN	operator requirements

Many organisations shape 6G. The ITU-R sets the frame. 3GPP specifies. The World Radiocommunication Conference decides on spectrum. Regional research programmes and industry alliances feed ideas into them, and the O-RAN ALLIANCE defines open interfaces inside the radio network.

ITU-R WP 5D · 3GPP · WRC-27

1.6 Lessons taken from 5G



The industry wants to avoid some 5G experiences. 5G was specified with many architecture options, which split the market and delayed standalone networks. For 6G the baseline is one option: standalone, with no dual connectivity to 5G. Other lessons are fewer optional features, energy saving from the start, and simple devices from the first release.

3GPP Release 20 6G study guidance · NGMN 6G position

1.7 Where things stand: October 2026

SNAPSHOT

3GPP Release 20	6G studies, running since mid-2025, ending in 2027
3GPP Release 21	first 6G specifications: stage 3 freeze planned for December 2028
ITU-R	requirements and evaluation method ready in 2026
WRC-27	October to November 2027: new IMT bands
Proposals to the ITU	2027 to early 2029
First networks	commercial launch expected around 2030

3GPP is in the study phase. The radio group has made first choices: OFDM waveform, reuse of the 5G channel codes, carriers from 3 to 400 MHz. The ITU has finished its requirements. The spectrum decision comes at WRC-27, the first 6G specifications at the end of 2028.

Status as of October 2026

1.8 Three things to remember

1

IMT-2030 = 6G

ITU sets the goals, 3GPP writes the standard.

2

Evolution in the radio, new in the capabilities

OFDM stays. Sensing, AI and satellites are added.

3

Dates: 2027, 2028, 2030

Spectrum, specifications, launch.

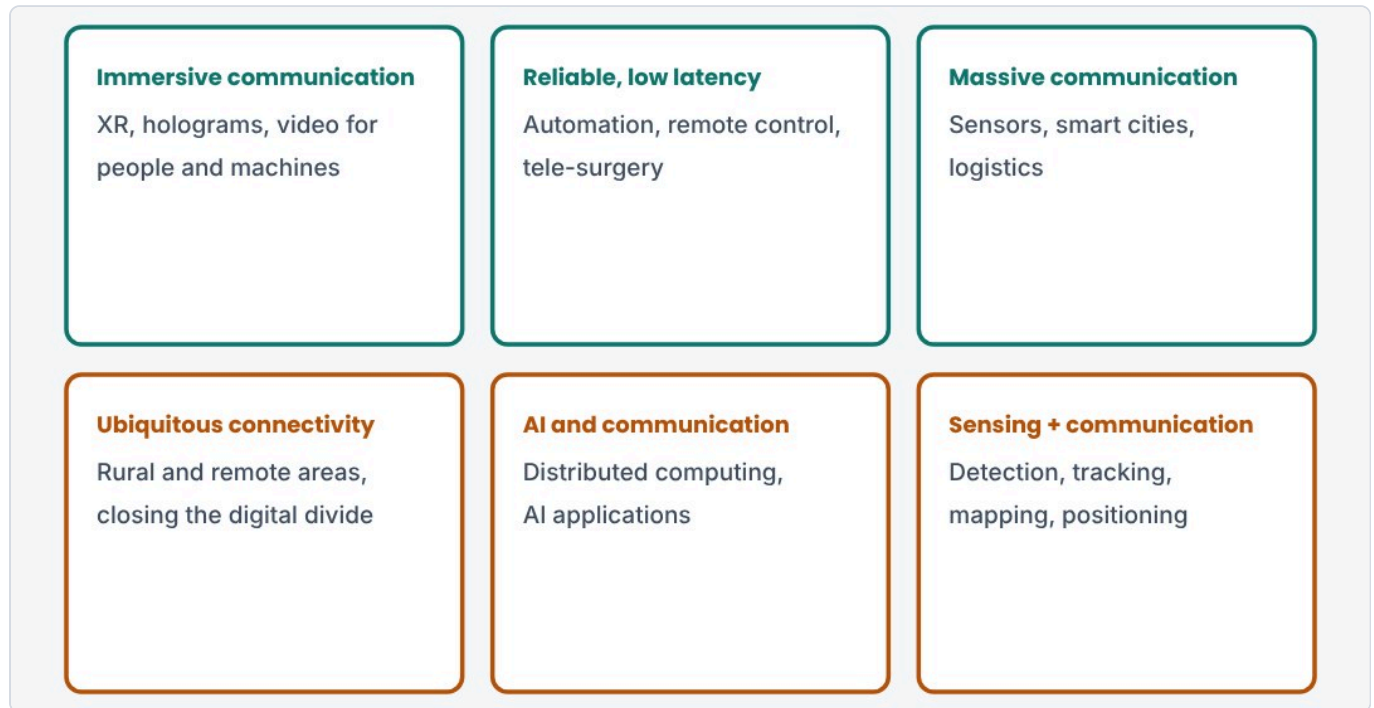
IMT-2030 is the requirement, 3GPP Release 21 the specification. 6G adds sensing, AI and ubiquitous coverage to better broadband. Nothing is final before the end of 2028.

ITU-R M.2160 · 3GPP planning

IMT-2030: Scenarios and Capabilities

What the ITU expects from 6G: six usage scenarios, the capability targets and how candidates are evaluated.

2.1 Six usage scenarios



IMT-2020 had three usage scenarios. IMT-2030 has six. Three extend the 5G ones: immersive communication, hyper reliable and low-latency communication, and massive communication. Three are new: ubiquitous connectivity, AI and communication, and integrated sensing and communication.

ITU-R M.2160 §4.2, Figure 2

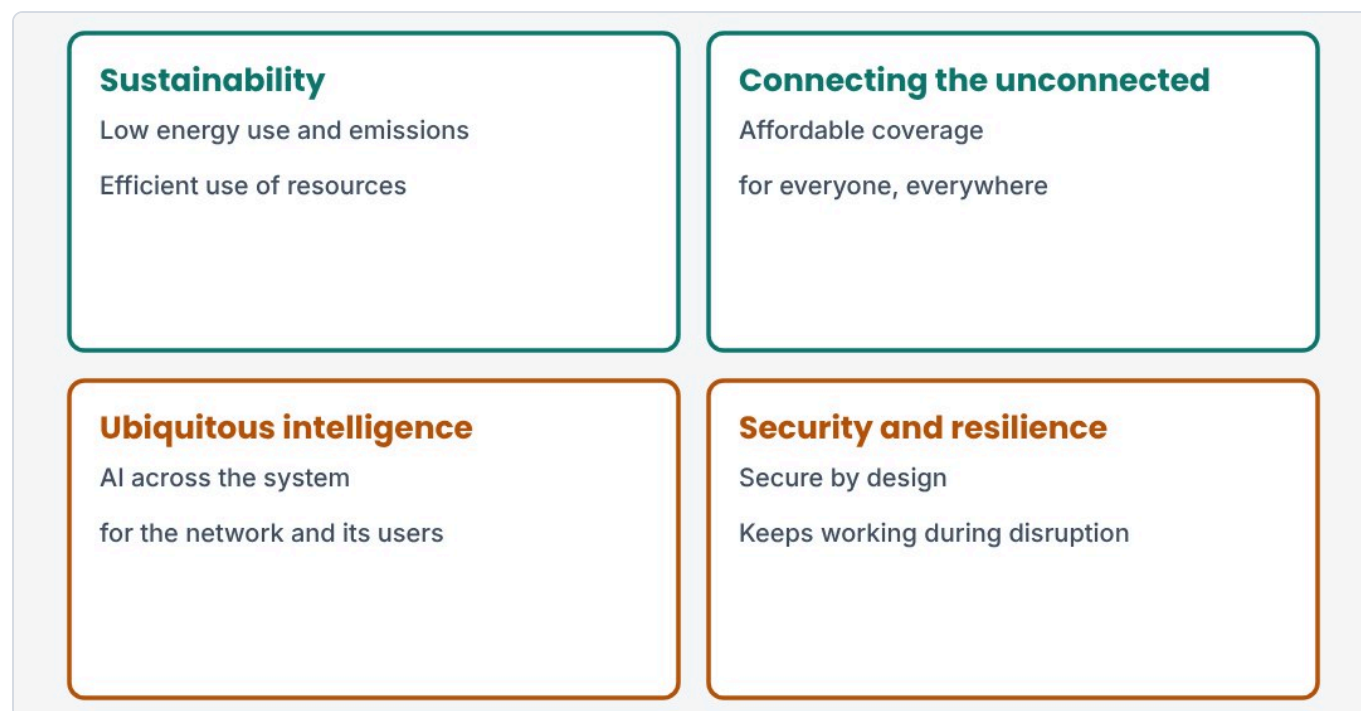
2.2 From the triangle to the hexagon

IMT-2020 (5G)	IMT-2030 (6G)
eMBB · mobile broadband	Immersive communication
URLLC · reliable, low latency	Hyper reliable and low-latency communication
mMTC · massive machines	Massive communication
—	Ubiquitous connectivity (new)
—	AI and communication (new)
—	Integrated sensing and communication (new)

The three IMT-2020 scenarios formed a triangle. Each corner now has a successor with stronger requirements, and three new scenarios sit between them. The new ones come from combining communication with something else: wider coverage, computing and AI, or sensing.

ITU-R M.2083 (IMT-2020) · ITU-R M.2160 (IMT-2030)

2.3 Four principles around the scenarios



The ITU places four overarching aspects around all six scenarios. They are design principles that apply to everything 6G does. They are not features to be added later but conditions every solution has to meet.

ITU-R M.2160 §4.2

2.4 Capabilities carried over from 5G

CAPABILITY	IMT-2030 RESEARCH TARGET	IMT-2020
Peak data rate	50, 100, 200 Gbit/s (examples)	20 Gbit/s
User experienced rate	300, 500 Mbit/s (examples)	100 Mbit/s
Spectrum efficiency	1.5 × and 3 × IMT-2020	1 ×
Area traffic capacity	30, 50 Mbit/s per m ²	10 Mbit/s per m ²
Connection density	10 ⁶ to 10 ⁸ devices per km ²	10 ⁶ per km ²
Mobility	500 to 1000 km/h	500 km/h
Latency	0.1 to 1 ms	1 ms
Reliability	1 – 10 ⁻⁵ to 1 – 10 ⁻⁷	1 – 10 ⁻⁵
Security and resilience	qualitative	–

Nine capabilities continue from IMT-2020 with higher goals. The ITU gives them as ranges or example values for research, not as fixed requirements. The values depend on the scenario: no single deployment must reach all of them at once.

ITU-R M.2160 §5, Figure 3, Table 1

2.5 Capabilities that are new

NEW IN IMT-2030

Coverage	reach of the service, including remote areas
Positioning	accuracy of 1 to 10 cm
Sensing	range, velocity and angle estimation, detection, imaging
Applicable AI	support for AI functions: distributed data, learning, inference
Sustainability	energy efficiency, low emissions, long equipment life
Interoperability	open interfaces, working across systems and vendors

Six capabilities are new in IMT-2030. Some have numbers, such as a positioning accuracy of 1 to 10 cm. Others are described in words, because agreed metrics do not exist yet. Defining how to measure them is part of the work.

ITU-R M.2160 §5

2.6 From capabilities to requirements

14 enhanced from IMT-2020 6 new positioning, sensing, AI ... 7 test environments	
TEST ENVIRONMENT	FOR SCENARIO
Indoor hotspot	immersive communication
Dense urban	immersive communication
Rural	immersive communication
Urban macro	massive communication
Indoor factory	hyper reliable, low latency
Indoor factory, urban macro	integrated sensing and communication

The framework gives directions. The next step turns them into technical performance requirements with exact values and test conditions. The ITU agreed 20 of them in 2026: 14 that continue from 5G in stronger form and 6 new ones. Each is checked in defined test environments.

ITU-R WP 5D: IMT-2030 technical performance requirements and evaluation (2026)

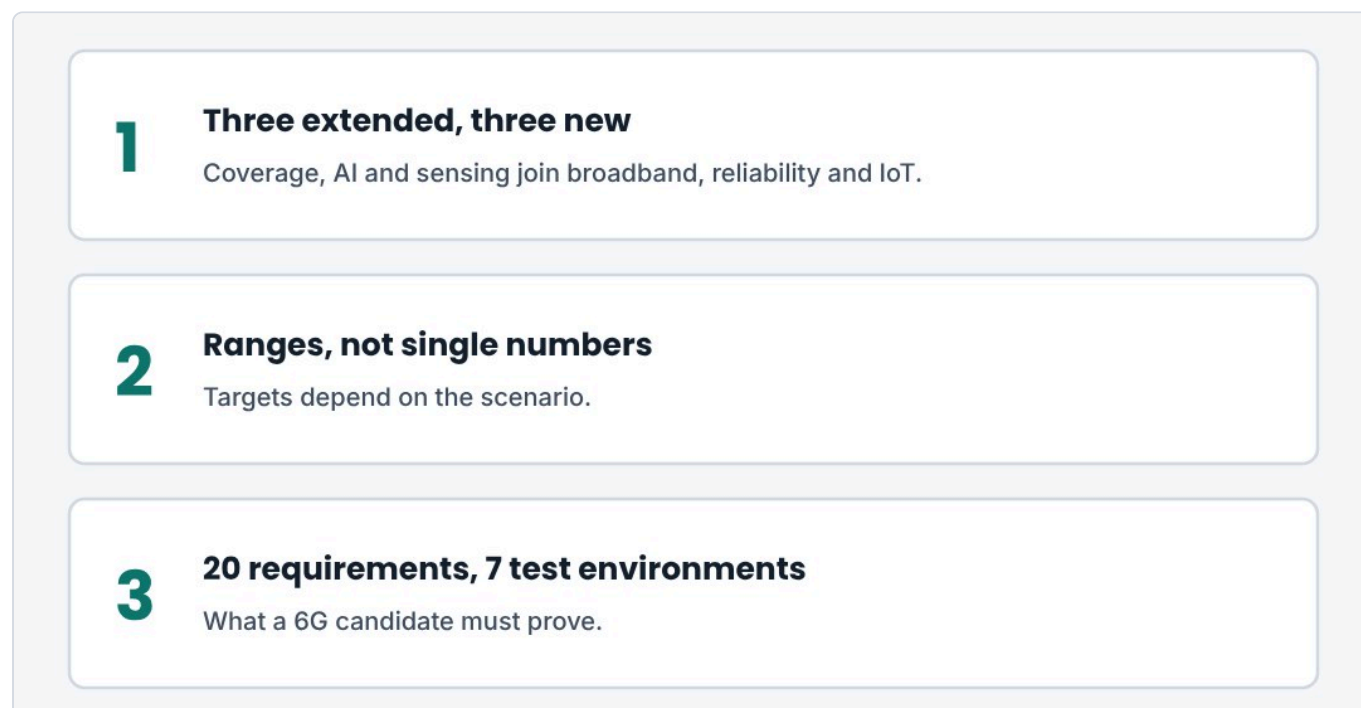
2.7 The IMT-2030 process



The ITU process runs in phases. The framework came in 2023 and the requirements in 2026. From 2027 standards bodies can submit candidate technologies. Independent groups evaluate them against the requirements. The detailed IMT-2030 specifications are planned for 2030.

ITU-R WP 5D IMT-2030 work plan

2.8 Three things to remember



Six scenarios, three of them new. Capabilities are research targets, not promises. The binding requirements were agreed in 2026.

ITU-R M.2160

Spectrum for 6G

Which frequencies 6G will use: existing bands, the new 7 to 15 GHz range, and sub-terahertz.

3.1 Layers of spectrum

LAYER	RANGE	ROLE
Low band	below 1 GHz	wide-area coverage, indoor reach
Mid band	1 to 7 GHz	main capacity layer of 5G, reused by 6G
Upper mid band ("FR3")	about 7 to 24 GHz	new capacity layer for 6G
Millimetre wave	24 to 71 GHz	hotspots, fixed wireless
Sub-terahertz	92 to 300 GHz	research: extreme rates, sensing, backhaul

"FR3" is an informal name: 3GPP has not defined it as a frequency range

No single band can do everything. Low bands travel far and enter buildings. Mid bands balance coverage and capacity. High bands offer wide channels over short distances. 6G will use all layers, and add a new one between the 5G mid band and millimetre wave.

3GPP TS 38.104 (FR1, FR2) · Release 20 6G studies

3.2 The bands on the WRC-27 agenda

BAND	STUDIED FOR
4400 – 4800 MHz	Regions 1 and 3 (parts)
7125 – 8400 MHz	Regions 2 and 3; 7125 – 7250 and 7750 – 8400 MHz in Region 1
14.8 – 15.35 GHz	all regions

Region 1: Europe, Africa, Middle East · Region 2: Americas · Region 3: Asia-Pacific

Nothing is decided before the end of 2027.

Administrations already disagree: some support only parts of these ranges.

Agenda item 1.7 of the 2027 World Radiocommunication Conference studies new bands for IMT. Three ranges are on the table. They are used today by satellites, military systems and fixed links, so sharing studies decide the outcome. The conference takes place in Shanghai from October 2027.

WRC-23 Resolution 256 · WRC-27 agenda item 1.7

3.3 Why 7 to 15 GHz

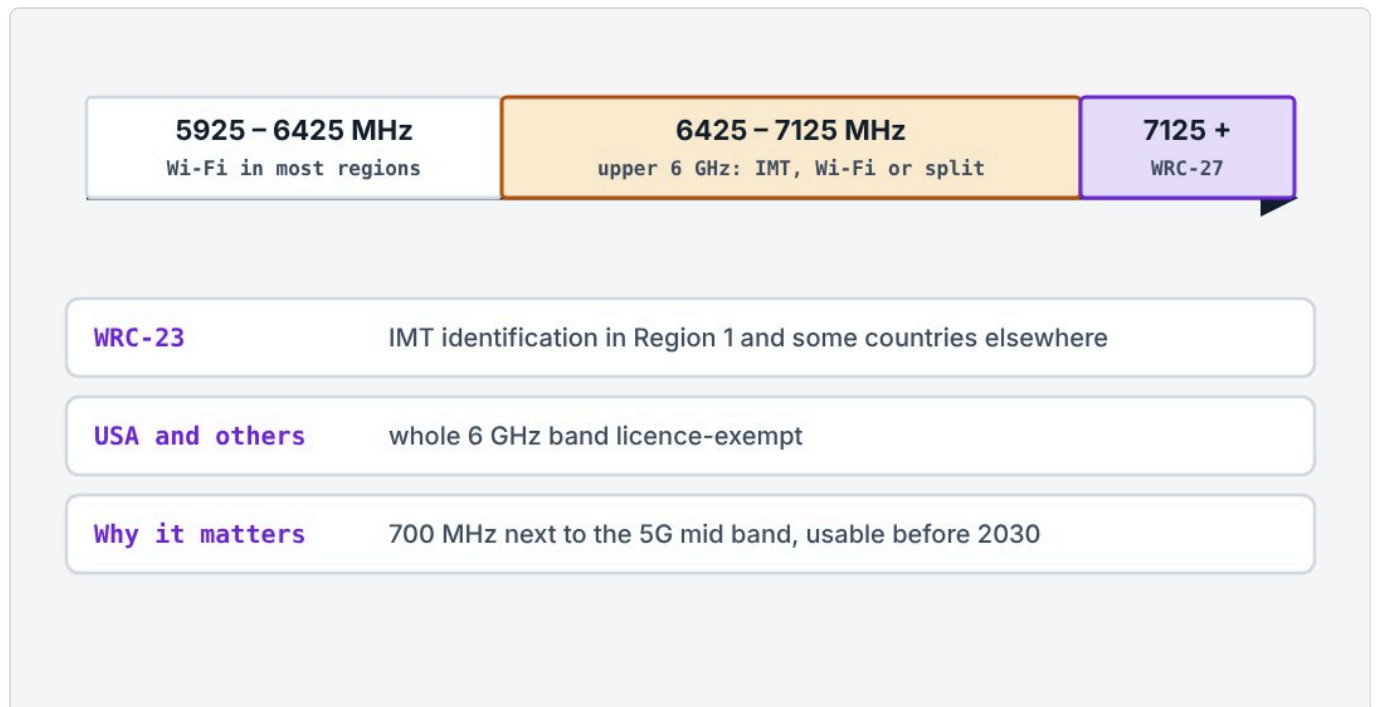
SAME ANTENNA SIZE, HIGHER FREQUENCY

Wavelength	3.5 GHz: 86 mm · 7 GHz: 43 mm · 15 GHz: 20 mm
Free-space loss	+6 dB for each doubling of the frequency
Elements per panel	× 4 for each doubling of the frequency
Array gain	+6 dB for × 4 elements, at the base station
Result	similar downlink reach to 3.5 GHz is the goal, on the same grid
Open issue	uplink and indoor coverage remain harder

This range has two advantages. There is room for carriers of several hundred megahertz. And the wavelength is short enough to fit many more antenna elements into the same panel size. The extra array gain can offset the higher path loss, so operators hope to reuse their existing mid-band sites.

3GPP Release 20 6G channel model and deployment studies

3.4 The upper 6 GHz band



The band from 6425 to 7125 MHz is the nearest new mid-band spectrum. WRC-23 identified it for IMT in Region 1 and in several countries of the other regions. Elsewhere it is used or planned for Wi-Fi. Regulators handle it differently, and some split the band between the two.

WRC-23 Final Acts, Resolution 220 · national decisions

3.5 Sub-terahertz

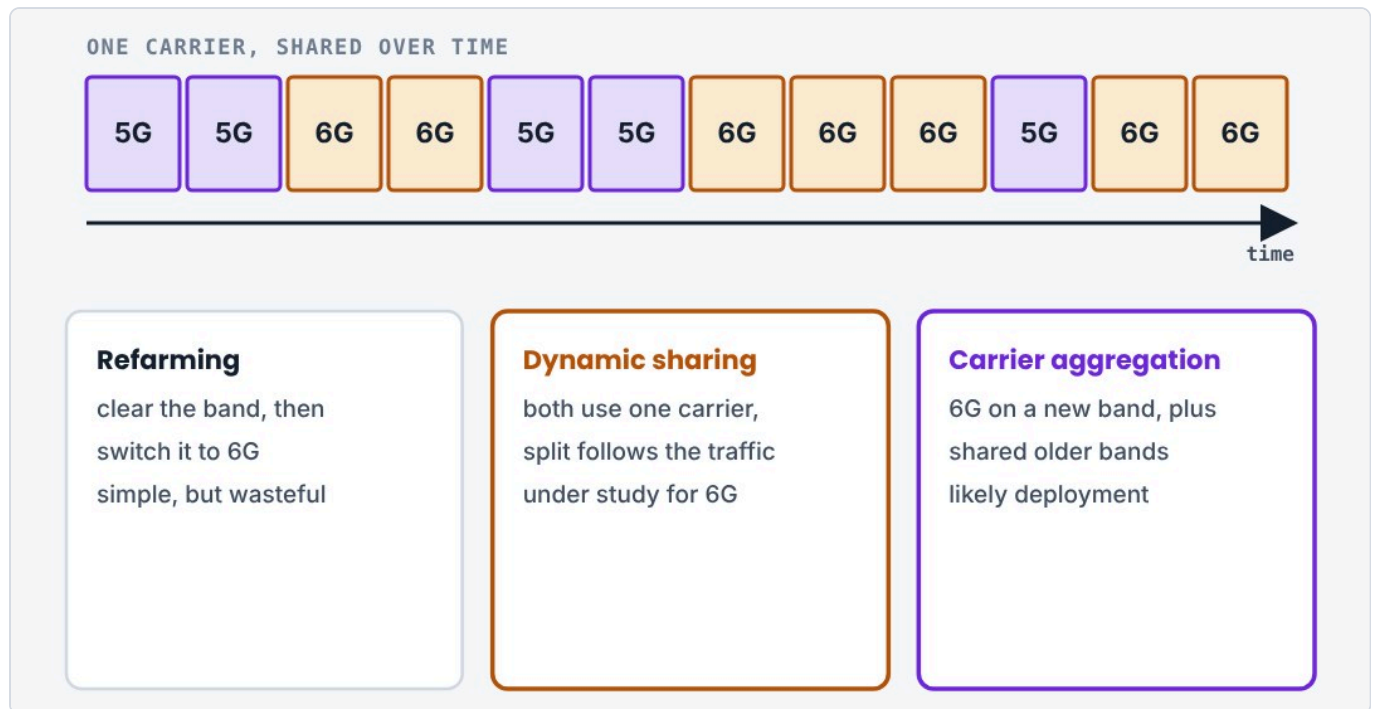
ABOVE 100 GHz

Bands	D band 110 – 170 GHz, and 275 – 450 GHz
WRC-19	137 GHz identified for land mobile and fixed use in 275 – 450 GHz
Strength	channels of several GHz, centimetre-level sensing
Weakness	high path loss, blockage, low transmit power
Likely uses	wireless backhaul and fronthaul, indoor hotspots, imaging
Timing	not part of the first 6G release

Above 100 GHz there are tens of gigahertz of spectrum. That allows very high rates and very fine sensing resolution. But the range is short, the signal is blocked easily and the hardware is immature. It is therefore expected in later 6G releases, for hotspots, backhaul and sensing, not for wide-area coverage.

ITU Radio Regulations footnote 5.564A (WRC-19) · ITU-R M.2541

3.6 Sharing spectrum with 5G



Most 6G traffic will at first run in bands that 5G uses today. Clearing a band for the new generation would waste capacity. 3GPP therefore studies how a 5G and a 6G carrier can share the same frequencies dynamically, with less overhead than the LTE and NR sharing of the past.

3GPP Release 20 6G study: multi-RAT spectrum sharing

3.7 Sharing with other services

INCUMBENT	TYPICAL CONCERN
Fixed satellite uplinks	aggregate interference into the satellite receiver
Earth exploration, astronomy	very sensitive passive receivers
Military systems	national security, often not public
Fixed links	co-channel interference on long hops
Aeronautical systems	safety of life services

A band identified for IMT may still come with tight technical conditions

Almost all candidate bands have existing users. Whether 6G can enter depends on coexistence studies. Typical measures are power limits, limits on emissions towards the sky, exclusion zones and databases that coordinate use. These conditions shape the real value of a band as much as its width.

ITU-R sharing and compatibility studies for WRC-27

3.8 Three things to remember

1	All layers, one new Low, mid, upper mid, millimetre wave.
2	4.4 – 4.8, 7.125 – 8.4, 14.8 – 15.35 GHz The ranges under study for WRC-27.
3	5G and 6G share Dynamic sharing instead of clearing bands.

6G starts in today's bands plus a new layer around 7 to 15 GHz. WRC-27 decides which parts. Sub-terahertz comes later.

WRC-27 agenda item 1.7

The 6G Air Interface

What has been agreed for the radio so far, and the research directions around antennas, duplexing and energy.

4.1 First decisions in 3GPP

ITEM	AGREEMENT SO FAR
Downlink waveform	CP-OFDM
Uplink waveform	CP-OFDM and DFT-s-OFDM, the latter now also with MIMO
Channel coding	5G codes (LDPC, Polar) largely reused, with enhancements
Modulation	uniform QAM as the baseline
Carrier bandwidth	3 MHz minimum, 400 MHz maximum
Frame structure	very similar to 5G
Operation	standalone as the baseline, no dual connectivity with 5G

By mid-2026 the radio group had fixed the basics. The waveform stays OFDM. The 5G channel codes are largely reused. Carriers can be from 3 to 400 MHz wide. These are working agreements of the study phase. They become binding only when Release 21 is written.

3GPP RAN 6G study, status June 2026 (public summaries)

4.2 Why OFDM stays

Reasons to keep OFDM

- Mature chips and algorithms
- Simple MIMO processing
- Flexible numerology
- Same grid as 5G: easy sharing

Alternatives that were studied

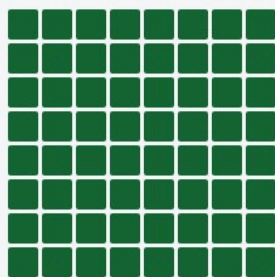
- OTFS: for very high Doppler
- Single-carrier variants: low PAPR
- Filtered and windowed OFDM
- Kept only as research topics

Evolution inside OFDM: MIMO for DFT-s-OFDM, wider carriers

Many new waveforms were proposed for 6G. OFDM was kept because it is well understood, works naturally with MIMO, and makes it easy to share a carrier with 5G. Its weaknesses are known and manageable. A new waveform would have had to be much better to justify the cost of change.

3GPP RAN1 6G waveform study

4.3 More antennas, larger arrays



5G, 3.5 GHz

64 transceivers, about 200 elements



6G, upper mid band

same size, over 1000 elements

Gain

narrower beams, higher array gain, more multi-user layers

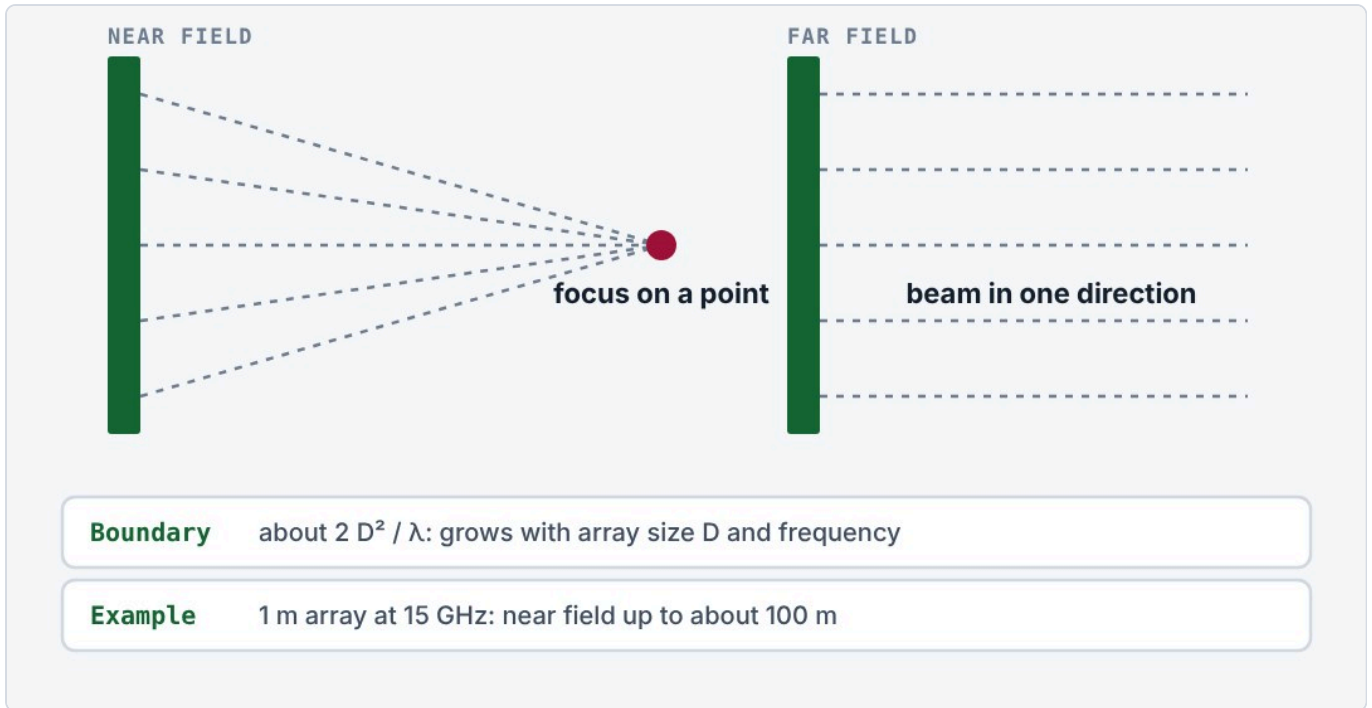
Cost

channel knowledge and feedback overhead grow with the array

The new upper mid band makes very large arrays practical. Where 5G base stations use 64 transceivers and a few hundred elements, 6G arrays may have more than a thousand elements. Beams get narrower, more users can be served on the same resources, and the array gain supports the higher frequency.

3GPP Release 20 6G MIMO study

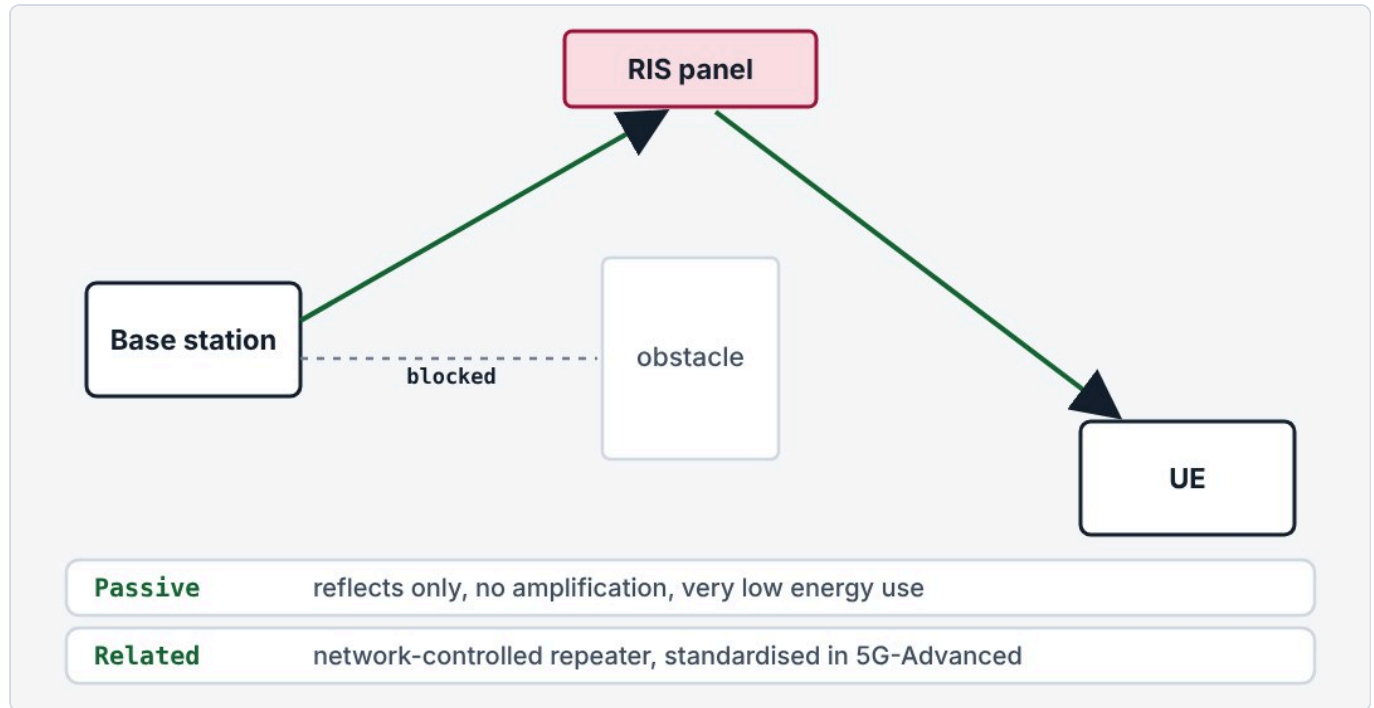
4.4 Near field and extra-large MIMO



With very large arrays and high frequencies, users can be inside the near field of the antenna. There the wavefront is curved, not flat. Beams can then be focused on a point in distance as well as direction. This helps to separate users but needs new channel models and codebooks.

3GPP TR 38.901 extensions (Release 19) · 6G channel model study

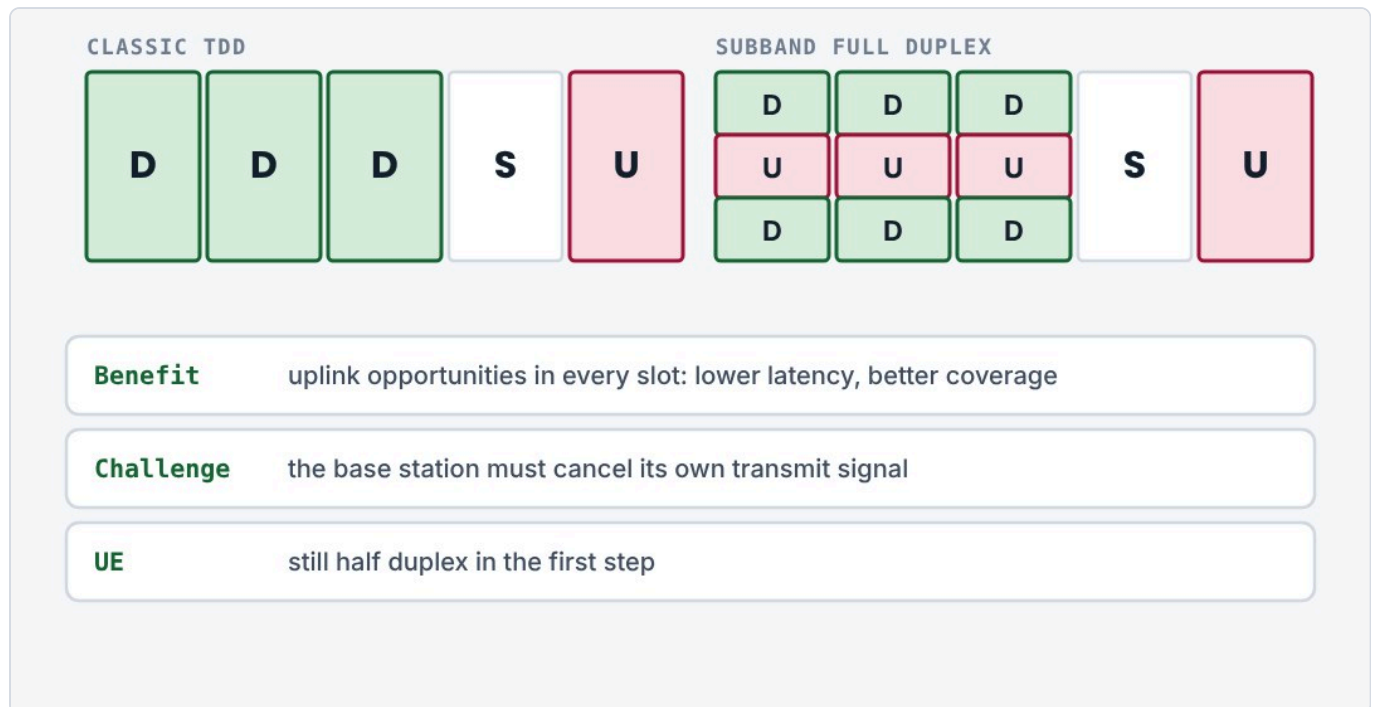
4.5 Reconfigurable intelligent surfaces



A reconfigurable intelligent surface is a panel of many small elements that reflect radio waves with a controllable phase. It can steer a reflection around an obstacle towards a user. It has no power amplifier, so it is cheap to run. Open questions are how to control it and how much it helps in real deployments.

ETSI ISG RIS reports · 3GPP network-controlled repeater (Release 18)

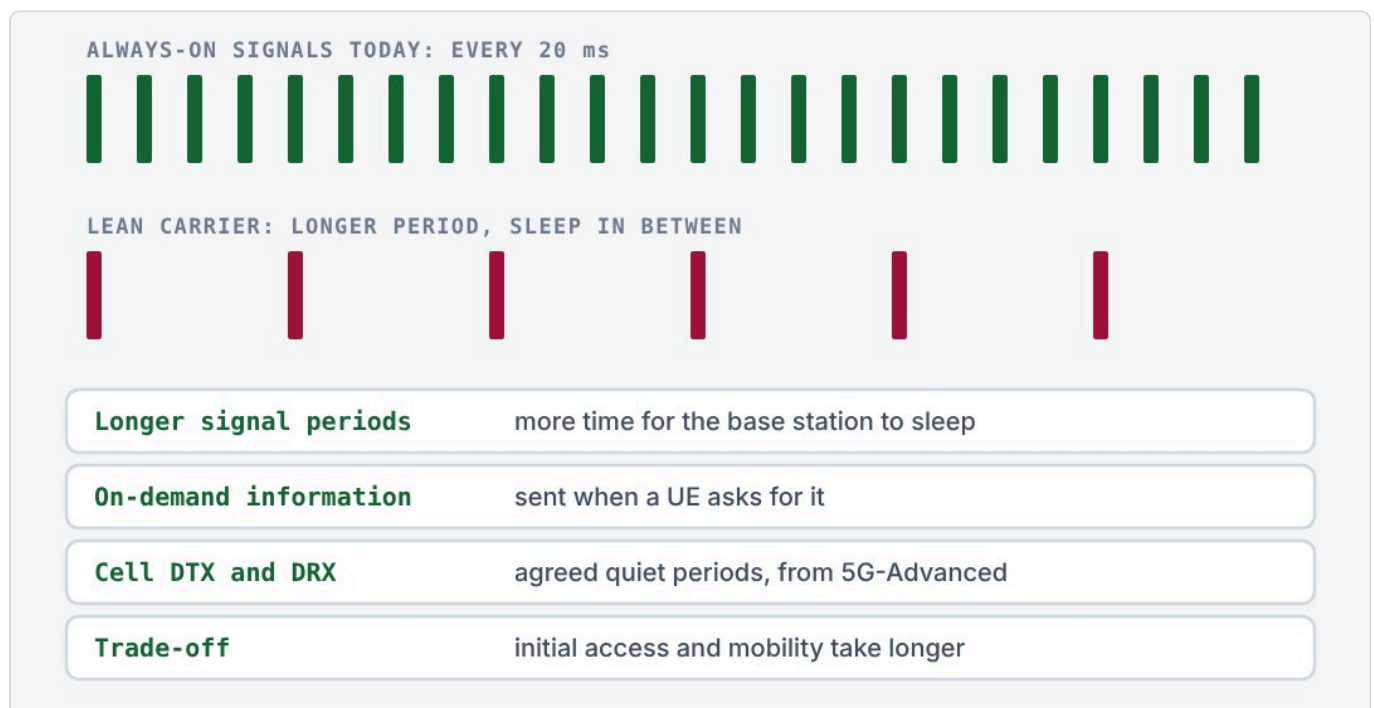
4.6 Duplex evolution



Today a TDD carrier is either in downlink or in uplink at a given time. Subband full duplex, introduced in 5G-Advanced, lets the base station receive in one part of the carrier while it transmits in the rest. This shortens uplink delay and improves coverage. 6G studies build on it.

3GPP TR 38.858 (duplex evolution) · Release 19 SBFD

4.7 Lean, energy-aware design



A 5G cell transmits reference and broadcast signals all the time, even with no users. 6G aims to send less when idle. Options under study are longer periods between synchronisation signals, system information on demand and deeper sleep modes for the base station.

3GPP TR 38.864 (network energy savings) · Release 20 6G study

4.8 Three things to remember

1

Evolution of the 5G radio

CP-OFDM, 5G channel codes, similar frame.

2

3 to 400 MHz carriers

Four times the 5G mid-band maximum.

3

Bigger arrays, leaner signals

Capacity up, idle energy down.

OFDM, LDPC and Polar continue. The gains come from wider carriers, larger arrays and smarter use of time and energy.

3GPP RAN 6G study

AI-Native Networks

What “AI-native” means for 6G: AI inside the radio, in the management of the network, and as a service the network supports.

5.1 Three roles of AI

AI for the network

- Better radio performance
- Automated operation
- Energy saving

The network for AI

- Moves data and models
- Offers compute close to users
- Predictable latency

AI as a service

- Models and inference offered to applications through network APIs

The first is furthest along: 5G-Advanced already standardises parts of it

AI appears in 6G in three ways. It improves the network itself, for example in beam management. The network in turn supports AI applications with data, computing and reliable transport. And AI functions may be offered to customers as a service. “AI-native” means these are designed in from the start and not added on top.

ITU-R M.2160 §4 · 3GPP TR 22.870

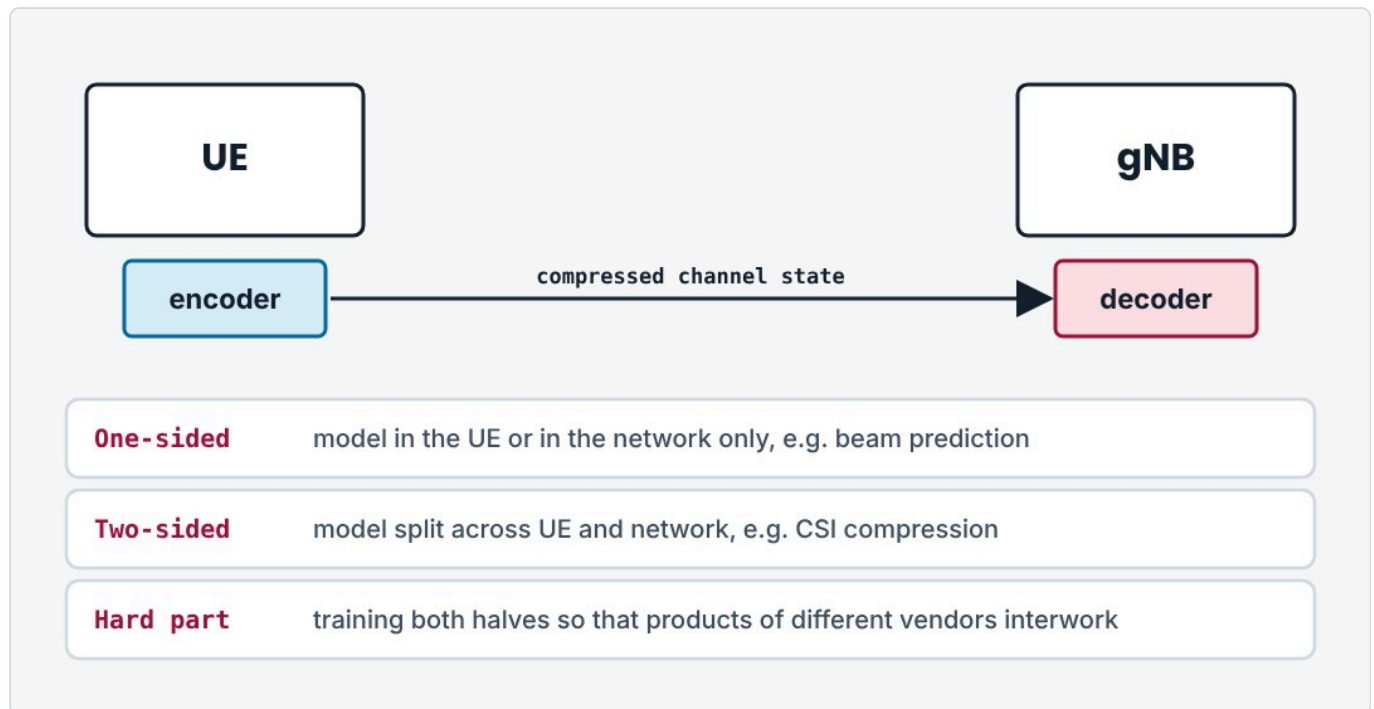
5.2 Where AI enters the radio

USE CASE	WHAT THE MODEL DOES
Beam management	predicts the best beam in space or time from few measurements
CSI feedback	compresses or predicts the channel state report
Positioning	estimates the position directly or improves the measurements
Mobility	predicts measurements and handover failures
Receiver	replaces or assists classic signal processing (research)

3GPP started with three use cases in the 5G-Advanced radio. A model predicts the best beam from a few measurements, compresses the channel state report, or improves positioning. Release 19 made the first of these normative. They are the starting point for the 6G work.

3GPP TR 38.843 (Release 18 study) · Release 19 AI/ML for NR air interface

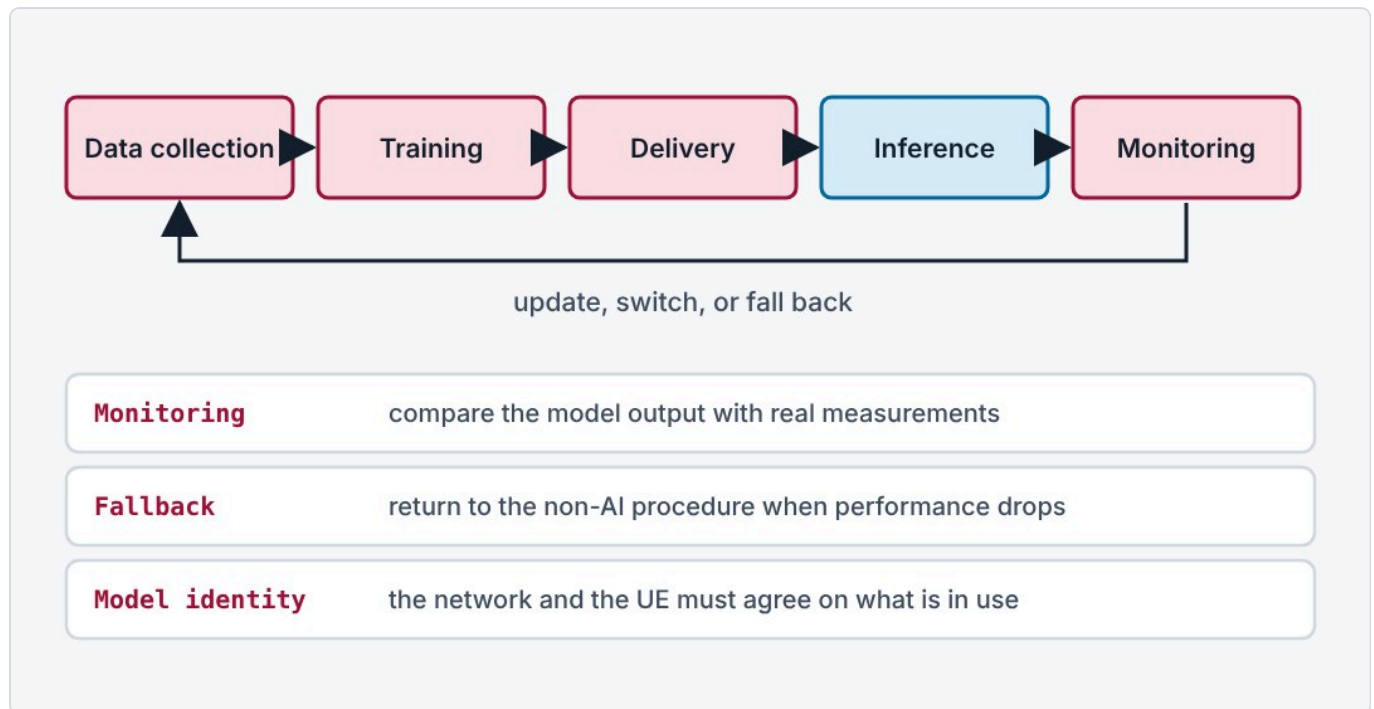
5.3 One-sided and two-sided models



Some models run in one place, in the UE or in the base station. They are the simpler case. Two-sided models are split: an encoder in the UE compresses information and a decoder in the network restores it. The two halves may come from different vendors, so they must be trained to work together.

3GPP TR 38.843 §4, §5

5.4 Life cycle management



A model is never finished. Data must be collected, the model trained, delivered, switched on and watched. When radio conditions change and the model gets worse, the system must notice and fall back to the classic method. Standardising this life cycle is the larger part of the AI work.

3GPP TR 38.843 §4.2 · Release 19 LCM framework

5.5 AI in the core and in operations

NWDAF

network data analytics function: load, mobility, QoS predictions

Intent-driven management

state the goal, the system derives the configuration

Closed loops

observe, decide, act, without a human in each step

RAN intelligent controller

O-RAN: xApps and rApps optimise the radio

AI agents

6G study topic: agents that plan and act across functions

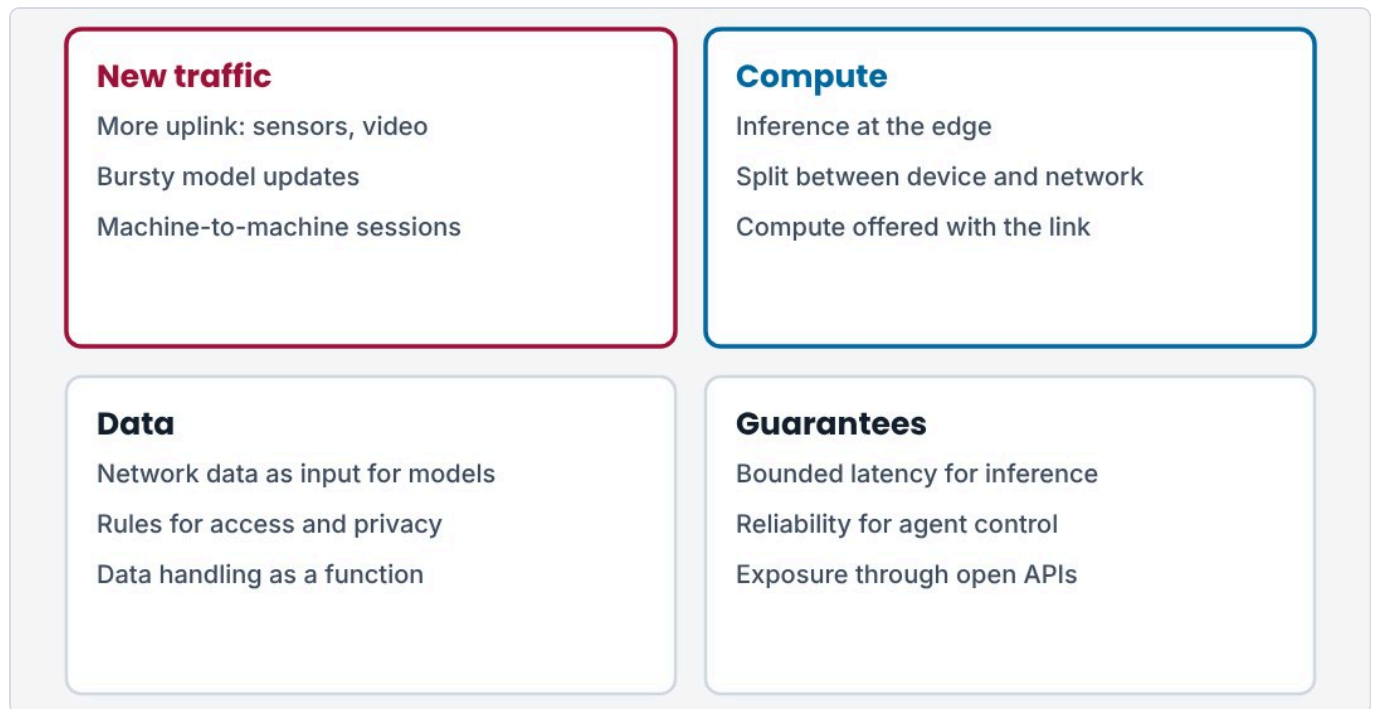
Digital twin

a virtual copy of the network to test changes safely

Beyond the radio, AI helps to run the network. In the 5G core the analytics function collects data and predicts load or mobility. Management systems move from scripts to intents: the operator states a goal and the system works out the configuration. 6G studies extend this towards AI agents.

3GPP TS 23.288 (NWDAF) · TS 28.312 (intent-driven management) · TR 22.870

5.6 The network as a platform for AI



AI applications also change what the network must deliver. Traffic becomes more uplink-heavy when devices send sensor data and video to models. Models and their updates must be moved around. Computing is needed close to the user. 6G treats computing and data as resources next to connectivity.

ITU-R M.2160 (AI and communication) · 3GPP TR 22.870

5.7 Open problems

PROBLEM	WHY IT IS HARD
Testing	behaviour depends on data, not only on the specification
Generalisation	a model trained in one cell may fail in another
Interoperability	two-sided models from different vendors
Energy	training and inference add power consumption
Trust	explainability, robustness against manipulated inputs
Data	who may collect it, and what it reveals about users

AI brings problems that classic radio features do not have. A model's behaviour depends on its training data, so testing it is hard. Results can differ between vendors. Training and inference cost energy. And decisions must be explainable enough for operators and regulators to trust them.

3GPP TR 38.843 §7 · RAN4 AI/ML testing work

5.8 Three things to remember

1

Beam, CSI, positioning

The first radio use cases, starting in Releases 18 and 19.

2

A model needs a life cycle

Collect, train, deliver, monitor, fall back.

3

The network becomes a compute platform

Connectivity, computing and data together.

AI for the network, the network for AI. The first steps are already in 5G-Advanced. Life cycle management and testing are the hard parts.

3GPP TR 38.843 · ITU-R M.2160

Integrated Sensing and Communication

Using the mobile network as a radar: how it works, what it can detect and what limits it.

6.1 The idea



A base station already sends radio signals in all directions. Those signals are reflected by cars, drones, people and buildings. If the network analyses the reflections, it can detect and track objects that carry no device at all. The same spectrum, the same hardware and the same signal serve both purposes.

3GPP TS 22.137 · ITU-R M.2160

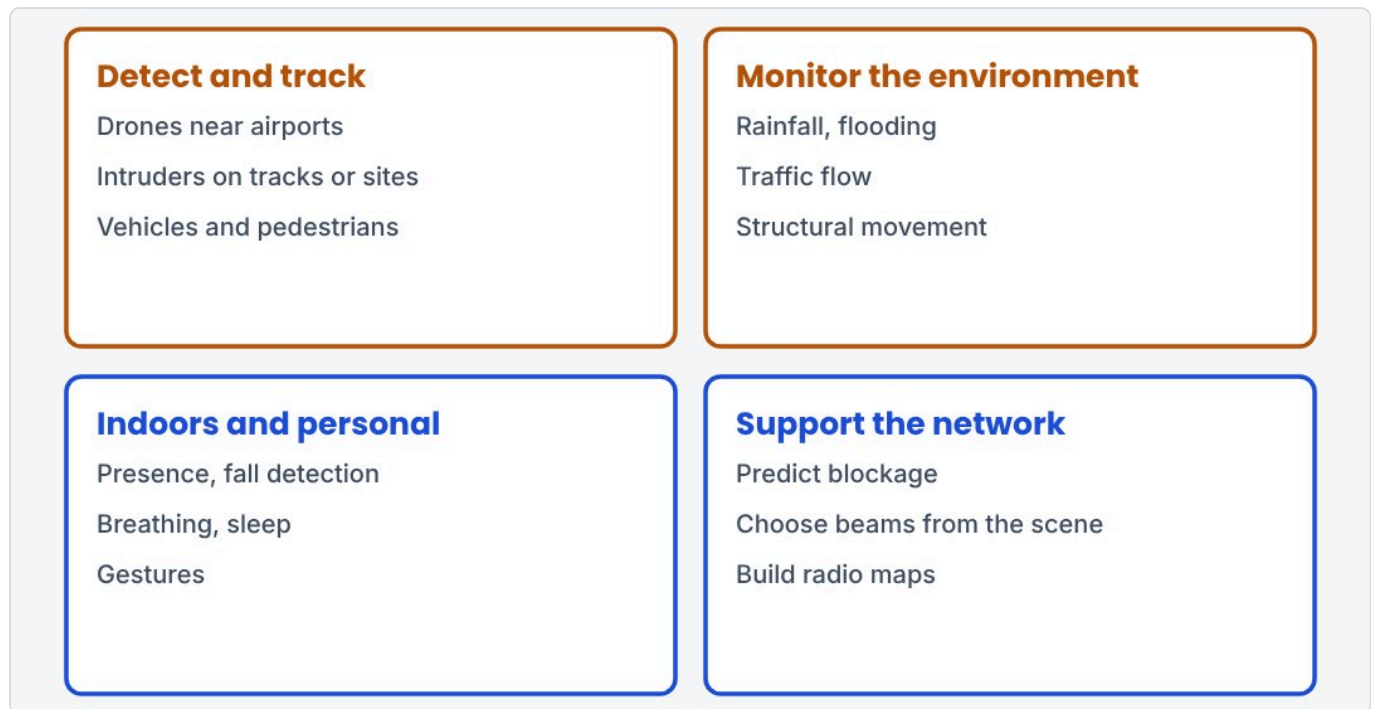
6.2 Six sensing modes

MODE	TRANSMITTER	RECEIVER
Base station monostatic	base station	same base station
Base station bistatic	base station A	base station B
Base station to UE	base station	UE
UE to base station	UE	base station
UE monostatic	UE	same UE
UE bistatic	UE A	UE B

Sensing is classified by who transmits and who receives. If the same node does both, it is monostatic. If they are different nodes, it is bistatic. With base stations and UEs in either role, 3GPP distinguishes six modes. The first 6G work concentrates on the ones that involve base stations.

3GPP TR 22.837 · Release 19 ISAC channel model study

6.3 What can be sensed



The requirements work collected dozens of use cases. They fall into a few groups: detecting intruders or drones, monitoring roads and railways, measuring weather, recognising gestures and monitoring health indoors. Each has its own needs for accuracy, range and update rate.

3GPP TR 22.837 · TS 22.137 §6

6.4 What limits the accuracy

QUANTITY	DEPENDS ON	FORMULA	EXAMPLE
Range resolution	bandwidth B	$c / (2B)$	100 MHz: 1.5 m
			400 MHz: 0.375 m
Velocity resolution	observation time T	$\lambda / (2T)$	20 ms: 0.5 m/s
Angle resolution	array aperture D	about λ / D	1 m at 15 GHz: 1.1°
Maximum range	power, gain, target	radar equation	hundreds of metres

Radar physics sets the limits. Range resolution depends on the bandwidth: wider carriers separate objects that are closer together. Velocity resolution depends on how long the object is observed. Angle resolution depends on the size of the antenna array. Wider 6G carriers and larger arrays help all three.

Radar fundamentals · 3GPP TS 22.137 KPIs

6.5 Reusing the communication signal



Delay → distance · Doppler → speed · array → direction

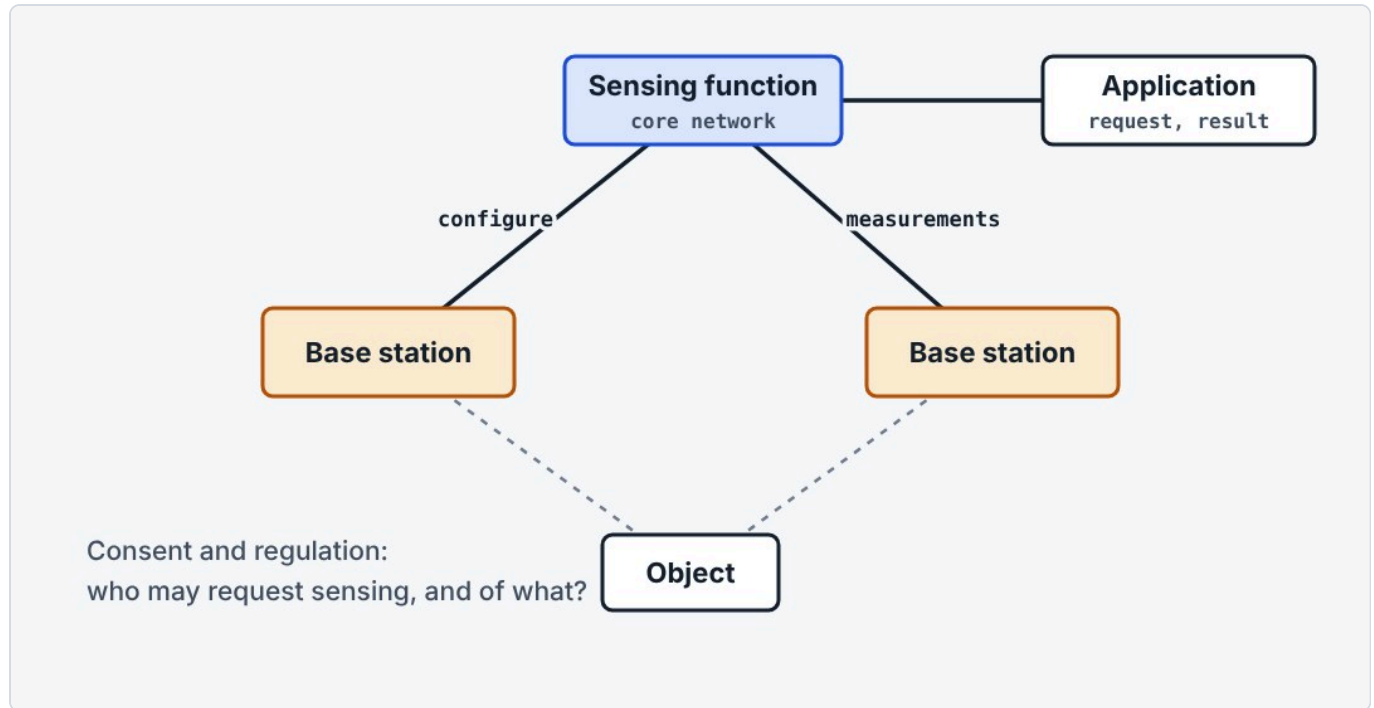
Reference signals or data symbols can be used

Dedicated sensing signals are an option when more accuracy is needed

OFDM is also a good radar waveform. The receiver knows the transmitted symbols, divides them out and is left with the channel. A delay shows up as a phase slope across subcarriers and a Doppler shift as a phase slope across symbols. Two FFTs turn the grid into a map of range and velocity.

OFDM radar processing · 3GPP Release 20 ISAC study

6.6 Sensing in the architecture



Sensing needs more than a radio feature. Something must accept a sensing request, pick the nodes that measure, collect the results and deliver them to the application. 3GPP studies a sensing function in the core for this. Privacy rules must cover people who are sensed without carrying a device.

3GPP TS 22.137 §5 · SA2 sensing study

6.7 Hard problems

PROBLEM	WHY
Self-interference	the echo is far weaker than the transmitted signal
Synchronisation	nanoseconds of timing error become metres of range error
Clutter	walls and ground reflect much more than a small drone
Resource sharing	sensing signals take time and frequency from data
Privacy	people are sensed without consent or device
Channel models	targets and their radar cross sections must be modelled

Several issues remain open. A base station that listens for echoes while transmitting must cancel its own strong signal. Bistatic sensing needs very accurate synchronisation between sites. Reflections from the static environment hide small targets. And sensing uses resources that data could have used.

3GPP Release 19 ISAC channel model · research literature

6.8 Three things to remember

1 Echoes carry information
Objects without any device can be detected.

2 Resolution = $c / (2B)$
400 MHz gives about 0.4 m.

3 A new function in the core
Requests, measurements, results, privacy.

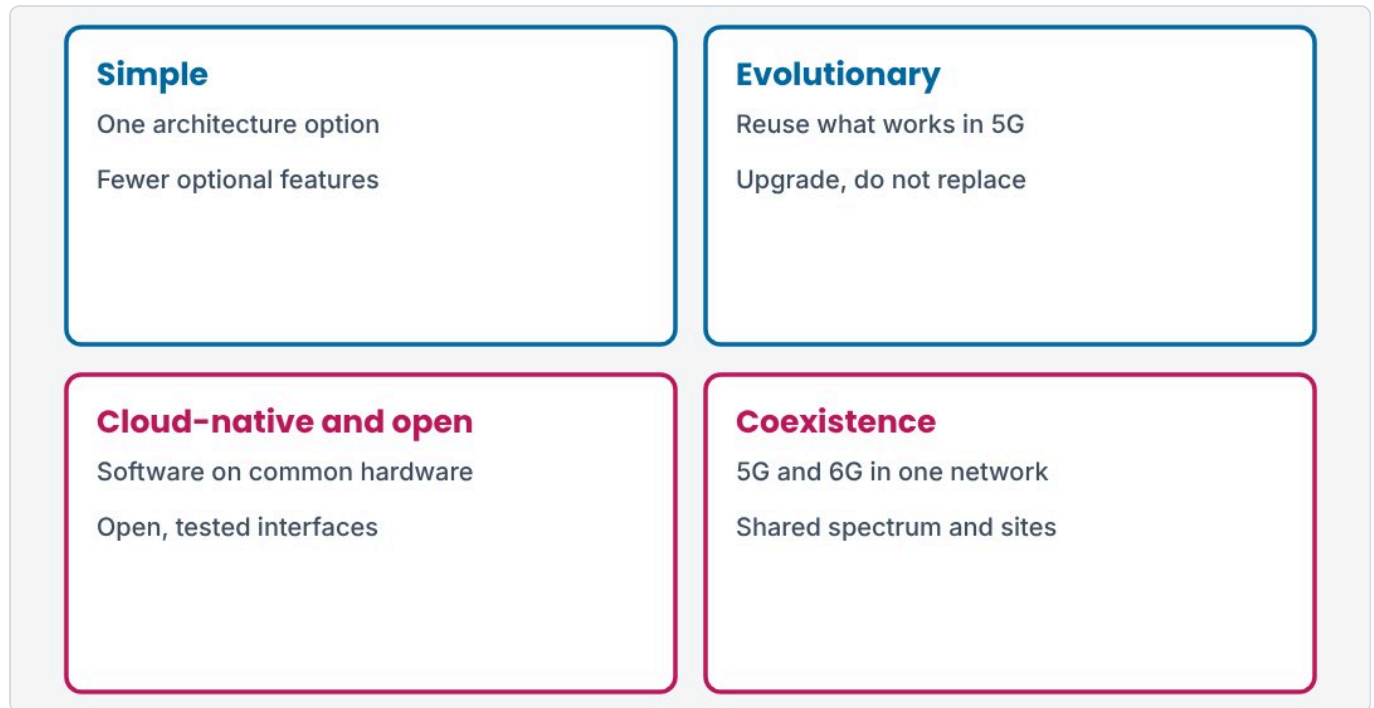
The network becomes a sensor. Bandwidth gives range resolution, the array gives direction. Privacy and self-interference are the open issues.

3GPP TS 22.137 · ITU-R M.2160

Architecture: RAN, Core and Cloud

How the 6G system is expected to be structured, what is carried over from 5G and how networks will migrate.

7.1 Design principles



The architecture work starts from principles more than from new boxes. Keep it simple, with one main deployment option. Build on what works in 5G. Make it cloud-native and open. Let 5G and 6G coexist smoothly. Most operators want an evolution they can upgrade into, not a full replacement.

3GPP SA and RAN 6G study guidance (Release 20) · NGMN

7.2 The many options of 5G, and the one of 6G

5G OPTION	RADIO	CORE
Option 3 (NSA)	LTE anchor + NR	EPC
Option 2 (SA)	NR	5GC
Option 4, 7	NR + LTE in various roles	5GC
Option 5	LTE	5GC

6G baseline: standalone

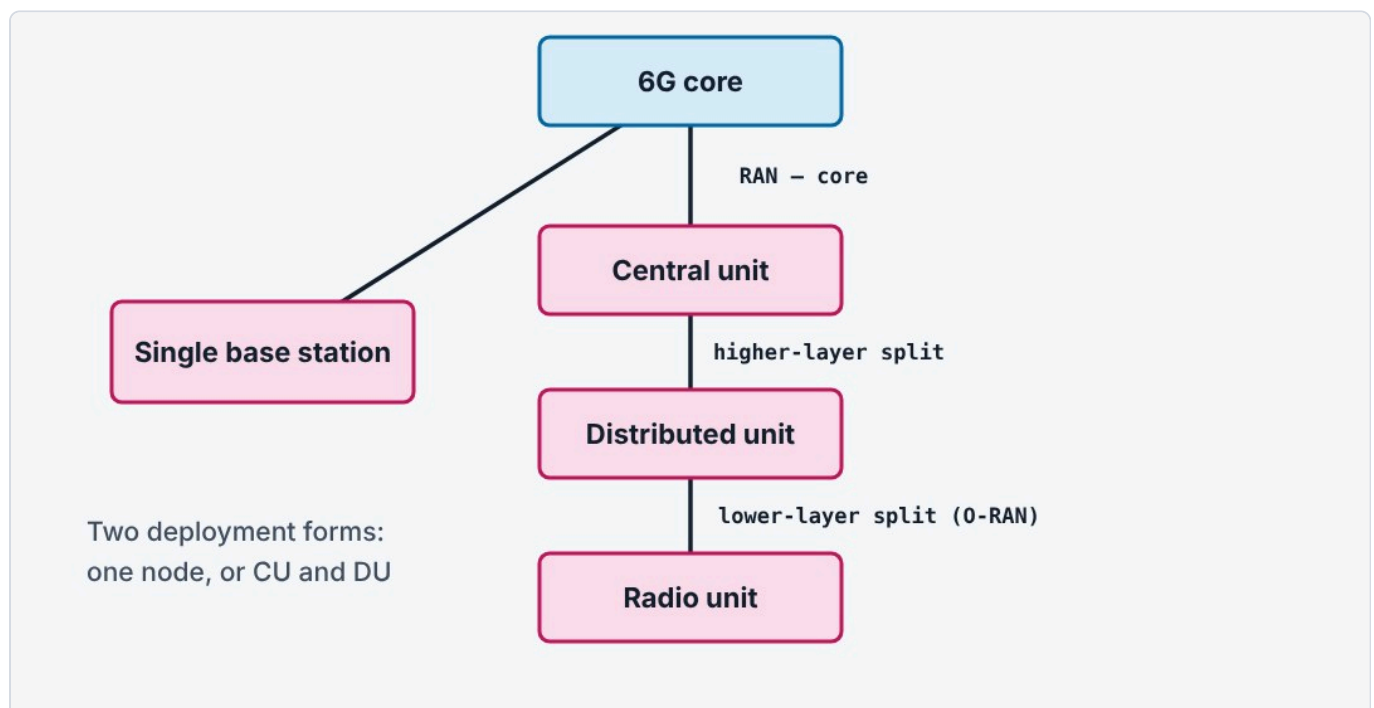
6G radio + 6G core. Migration by spectrum sharing, not by dual connectivity.

Status of the study: mid-2026

5G defined several ways to combine new and old radio and core. Most networks started non-standalone, with LTE as the anchor, and many took years to move on. The 6G baseline is standalone only: a 6G radio connected to a 6G-capable core, without dual connectivity to 5G.

3GPP TR 38.801 (5G options) · Release 20 6G RAN study

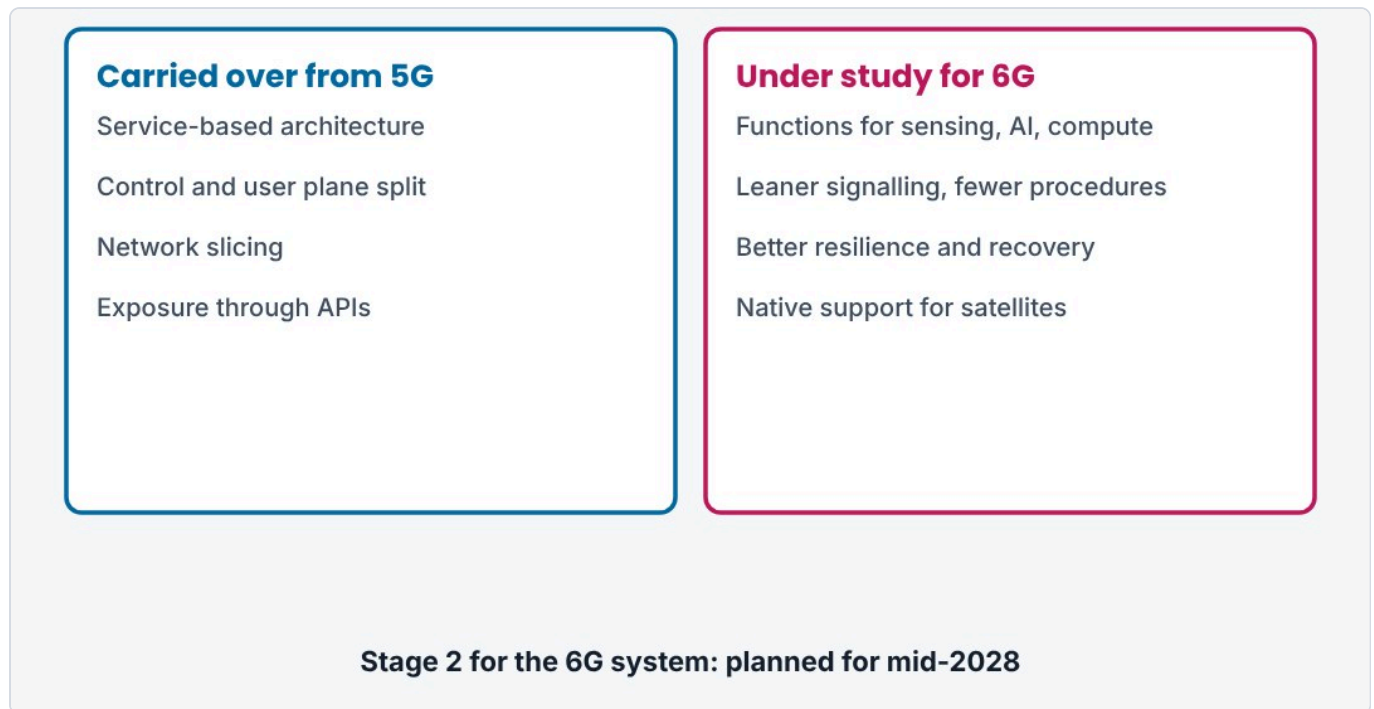
7.3 The radio access network



The 6G base station keeps the structure known from 5G. It can be one unit or be split into a central and a distributed unit. The interface towards the core stays a point-to-point interface. The split inside the radio unit, the lower-layer split, is left to the O-RAN ALLIANCE.

3GPP RAN3 6G study (2026) · O-RAN ALLIANCE

7.4 The core network



For the core the main question is how much changes. The 5G core is already service-based and cloud-native. The study looks at how to extend it for 6G: new functions for sensing, computing and AI, simpler procedures and better resilience. A completely new core is not the starting assumption.

3GPP SA2 6G architecture study (Release 20) · TS 23.501

7.5 Migration from 5G

STEP BY STEP

Today

5G standalone on all bands

6G start

6G on a new band, sharing some 5G carriers

Later

most bands shared or moved to 6G

Reused

sites, antennas where bands match, transport, cloud platform

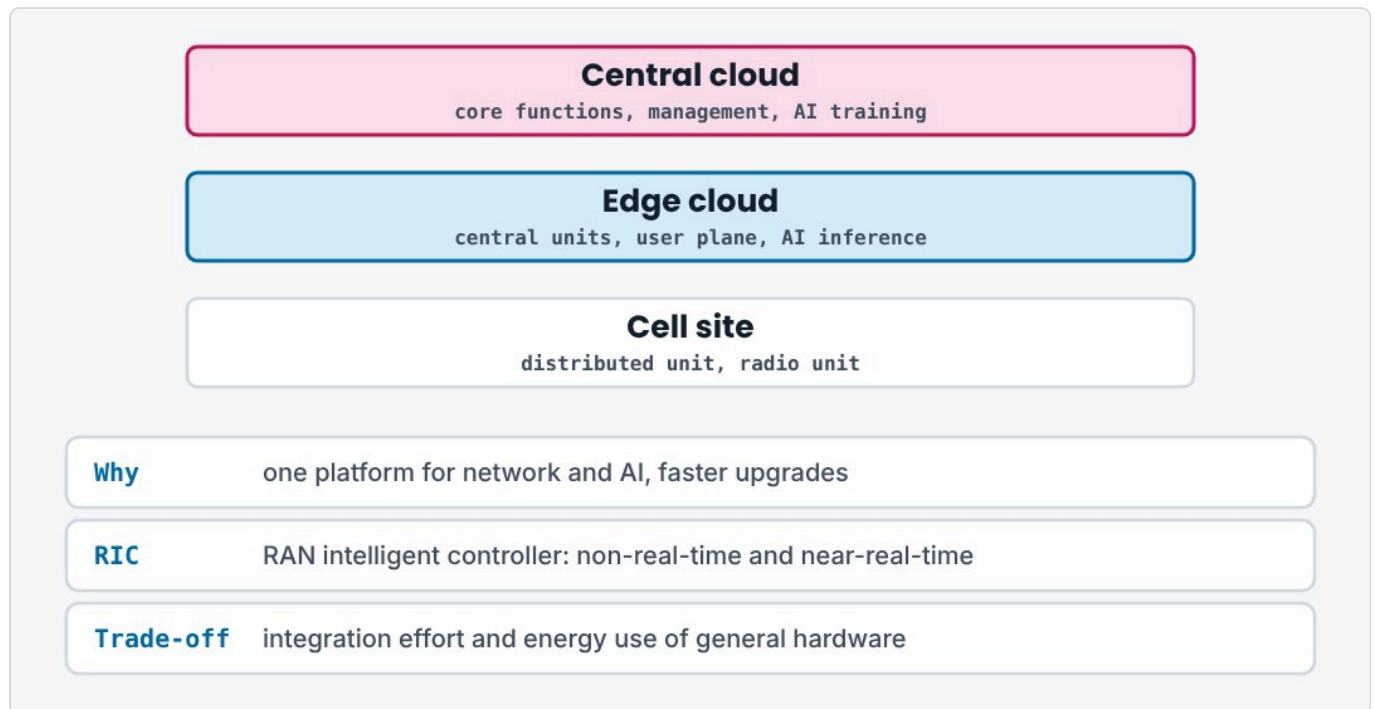
New

radio software or hardware, core functions, devices

Operators will introduce 6G into running 5G networks. Since dual connectivity is not the baseline, the tool is spectrum sharing: 5G and 6G use the same carrier and the split follows the traffic. Sites, transport and much of the core software can be reused.

3GPP Release 20 6G study: migration and spectrum sharing

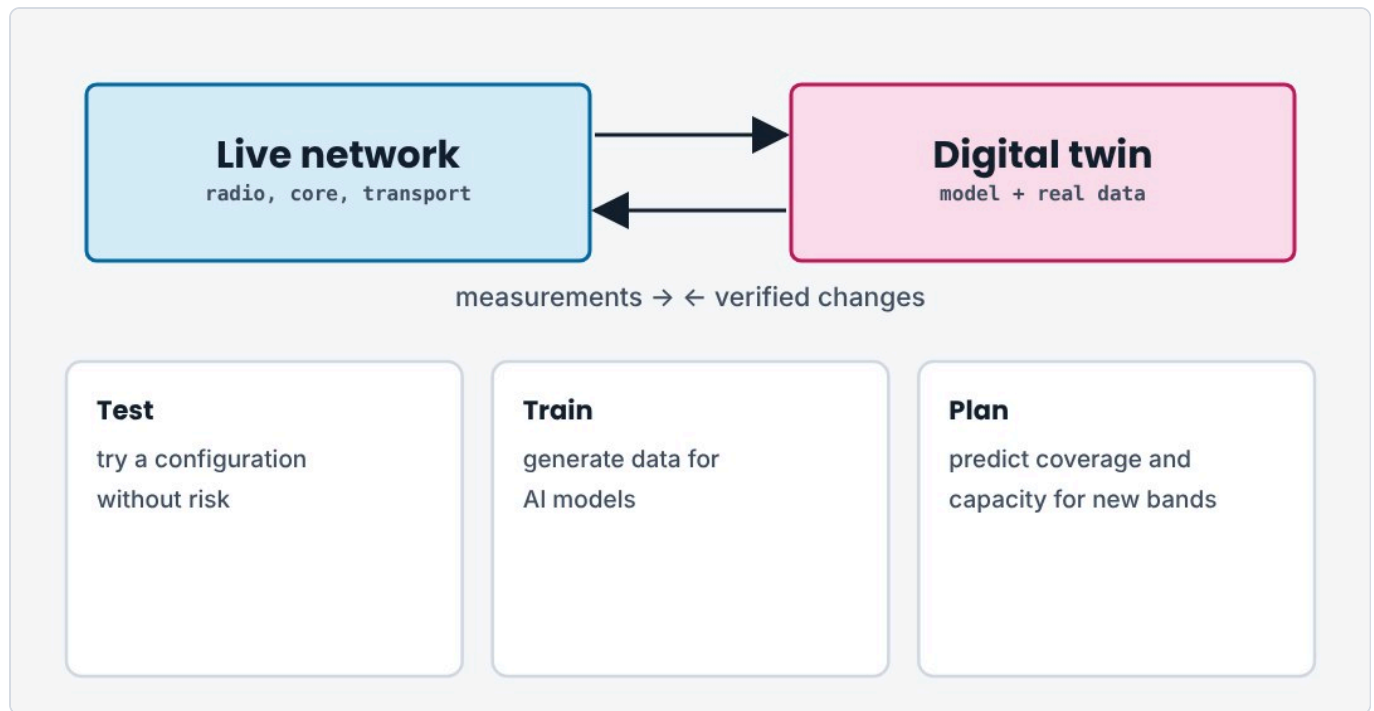
7.6 Cloud, edge and open RAN



6G assumes that network functions run as software on shared cloud infrastructure, from large data centres to small edge sites. Open interfaces let an operator mix vendors. An intelligent controller hosts applications that tune the radio. The same infrastructure can host AI workloads.

0-RAN ALLIANCE architecture · ETSI NFV · 3GPP TS 28 series

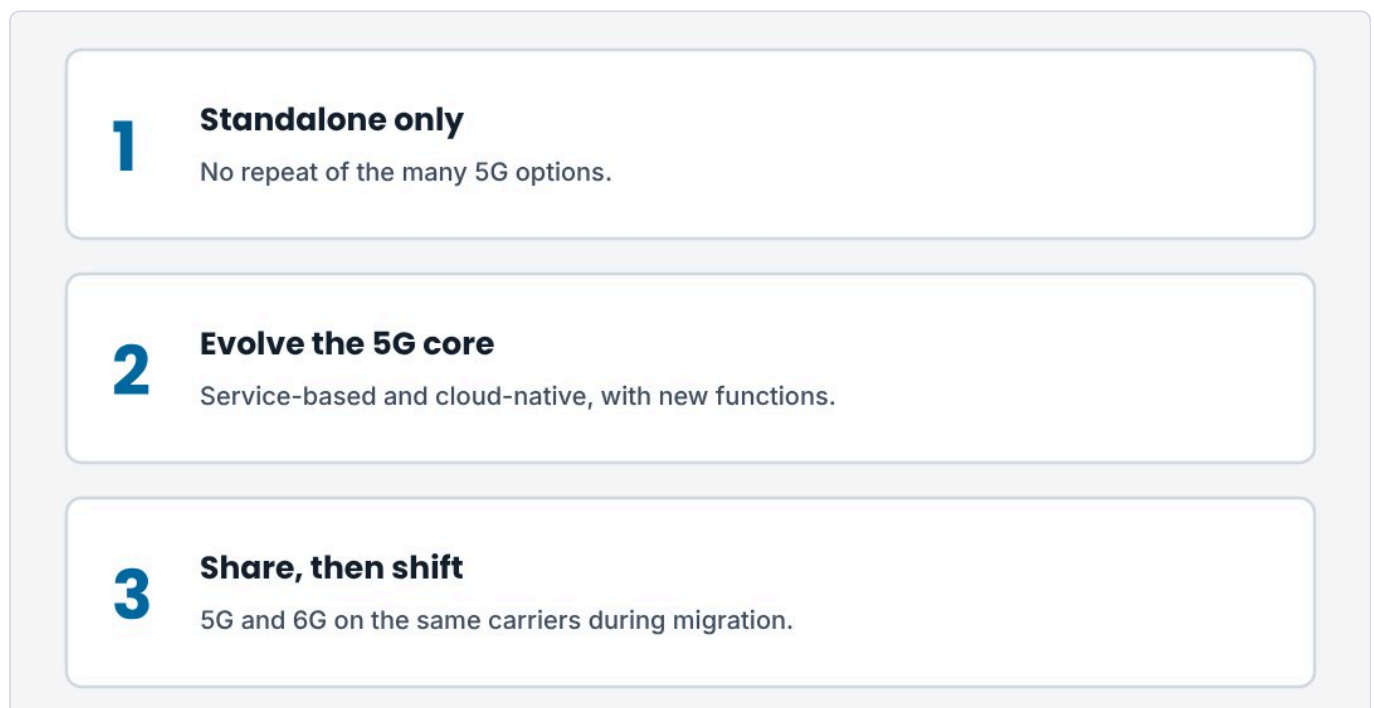
7.7 Network digital twins



A digital twin is a software model of the live network, fed with real data. Changes can be tried on the twin before they reach customers. It also produces data to train AI models and helps to plan coverage for new bands. 6G studies treat it as a tool for operation.

ITU-T Y.3090 · 3GPP TR 28.915 · 3GPP TR 22.870

7.8 Three things to remember



One architecture option, standalone. The 5G structures continue in RAN and core. Migration relies on sharing spectrum with 5G.

3GPP Release 20 6G studies

Non-Terrestrial Networks

Satellites and high-altitude platforms as part of the mobile network: orbits, payload types and the radio challenges.

8.1 Layers above the ground

PLATFORM	ALTITUDE	ONE-WAY DELAY	MOVES?
GEO satellite	35 786 km	about 120 ms	fixed in the sky
MEO satellite	7000 – 25 000 km	25 – 85 ms	slowly
LEO satellite	300 – 1500 km	1 – 5 ms	7.5 km/s
HAPS	about 20 km	below 0.1 ms	nearly fixed
Terrestrial cell	tens of metres	microseconds	no

Delay: UE to platform. A bent-pipe link doubles it.

A non-terrestrial network uses platforms in the sky as base stations or relays. Geostationary satellites cover a third of the Earth each but are far away. Low Earth orbit satellites are close and fast and need large constellations. High-altitude platforms fly in the stratosphere over one region.

3GPP TR 38.811 · TR 38.821

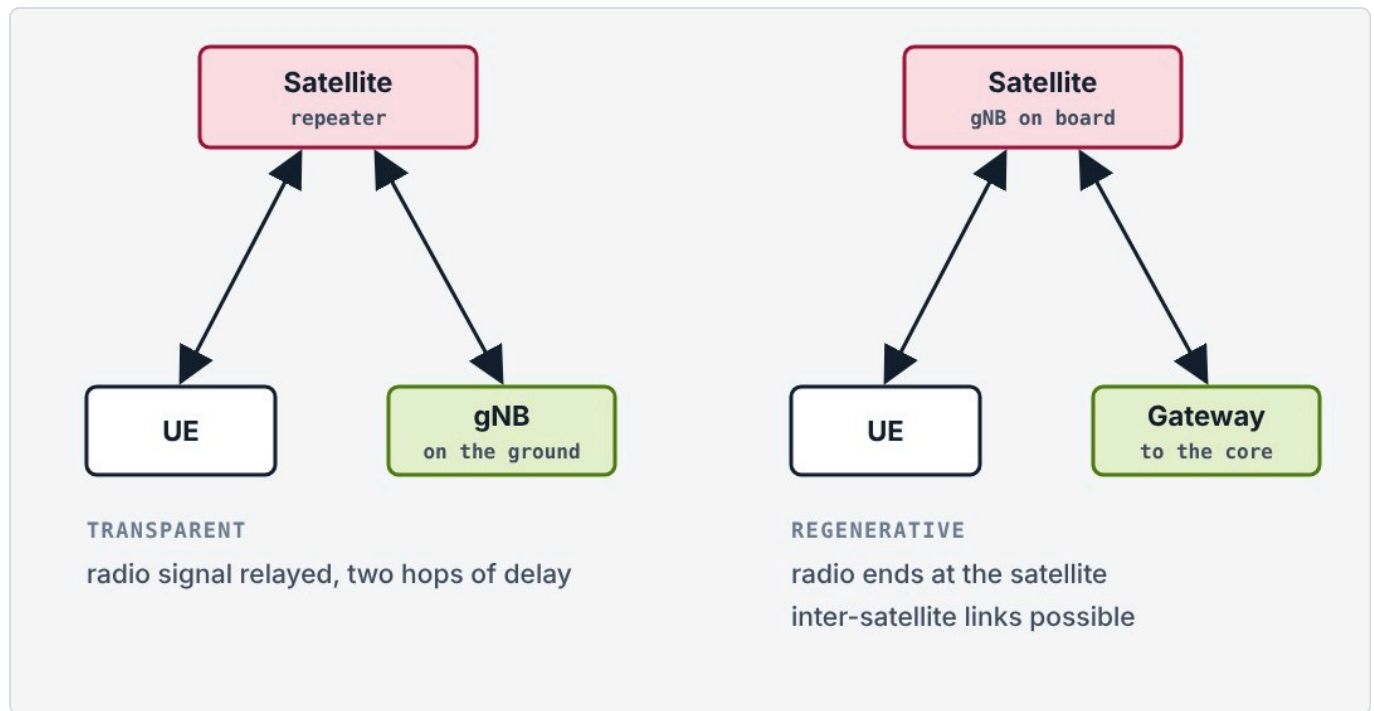
8.2 From add-on to native

RELEASE	STEP
15, 16	studies: channel models, scenarios, solutions
17	first specifications: NR over satellite, IoT over satellite
18	coverage, mobility, bands above 10 GHz
19	regenerative payload, more uplink capacity, RedCap over NTN
20	5G-Advanced enhancements, 6G NTN study
21	6G with terrestrial and non-terrestrial access in one design

Satellite support entered 3GPP step by step. Release 17 adapted NR and NB-IoT to work over satellites. Releases 18 and 19 improved coverage and mobility and added satellites that carry a full base station. For 6G the goal is one design that covers ground and space from the first release.

3GPP Release 17 NR-NTN and IoT-NTN · Releases 18, 19 · Release 20 6G study

8.3 Transparent and regenerative payloads



A transparent satellite only shifts and amplifies the signal: the base station stays on the ground. A regenerative satellite carries base station functions on board. That removes one leg of delay, allows links between satellites and lets the satellite serve areas where no ground station is in view.

3GPP TR 38.821 §4 · Release 19 regenerative payload

8.4 What is hard about satellite radio

Delay	LEO at 600 km: 4 to 13 ms round trip. GEO: about 500 ms through the gateway
Fix	UE uses its position and the satellite orbit data to pre-compensate timing
Doppler	LEO: up to ± 24 ppm, about ± 48 kHz at 2 GHz
Fix	UE pre-compensates the frequency with the same data
Moving cells	a LEO beam passes a point in a few minutes
Fix	time- and location-based handover, group handover

Three effects dominate. The delay is far larger than any terrestrial cell and differs across the beam. A low orbit satellite moves so fast that the Doppler shift is tens of kilohertz. And the cells move over the ground, so every UE is handed over every few minutes even when it stands still.

3GPP TR 38.821 §6 · TS 38.213 §4.2 (NTN timing)

8.5 Direct to device

In satellite bands (MSS)

L band and S band, e.g. n255, n256

Standard 3GPP NTN

Needs the band in the phone

In terrestrial mobile bands

Operator spectrum reused from space

Works with existing phones

Needs regulatory approval

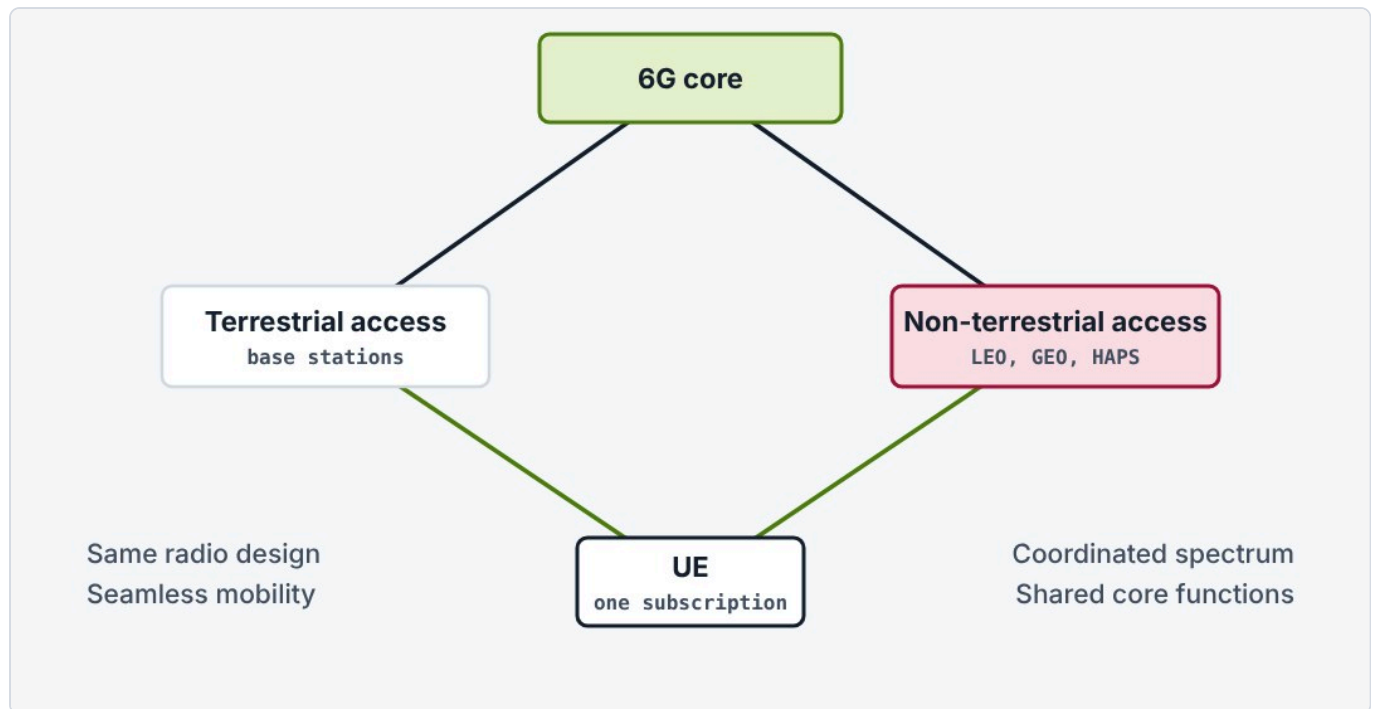
Limit phone antenna and 23 dBm against hundreds of kilometres

Services text first, then voice and moderate data

The most visible use is a normal phone that connects to a satellite when no ground network is available. It can use satellite bands or, with agreement of the regulator and operator, terrestrial mobile bands. Link budgets are tight, so the first services are messaging and low-rate data, with voice and broadband following.

3GPP Release 17 to 19 NTN bands (n255, n256 ...) · national regulations

8.6 One network, ground and space



In 6G the satellite part should not be a separate system. The same radio design, the same core and the same subscription cover both. A device moves between terrestrial and satellite cells like between two normal cells. Planning, spectrum use and interference must then be coordinated across both.

3GPP Release 20 6G NTN study · ITU-R M.2160 (ubiquitous connectivity)

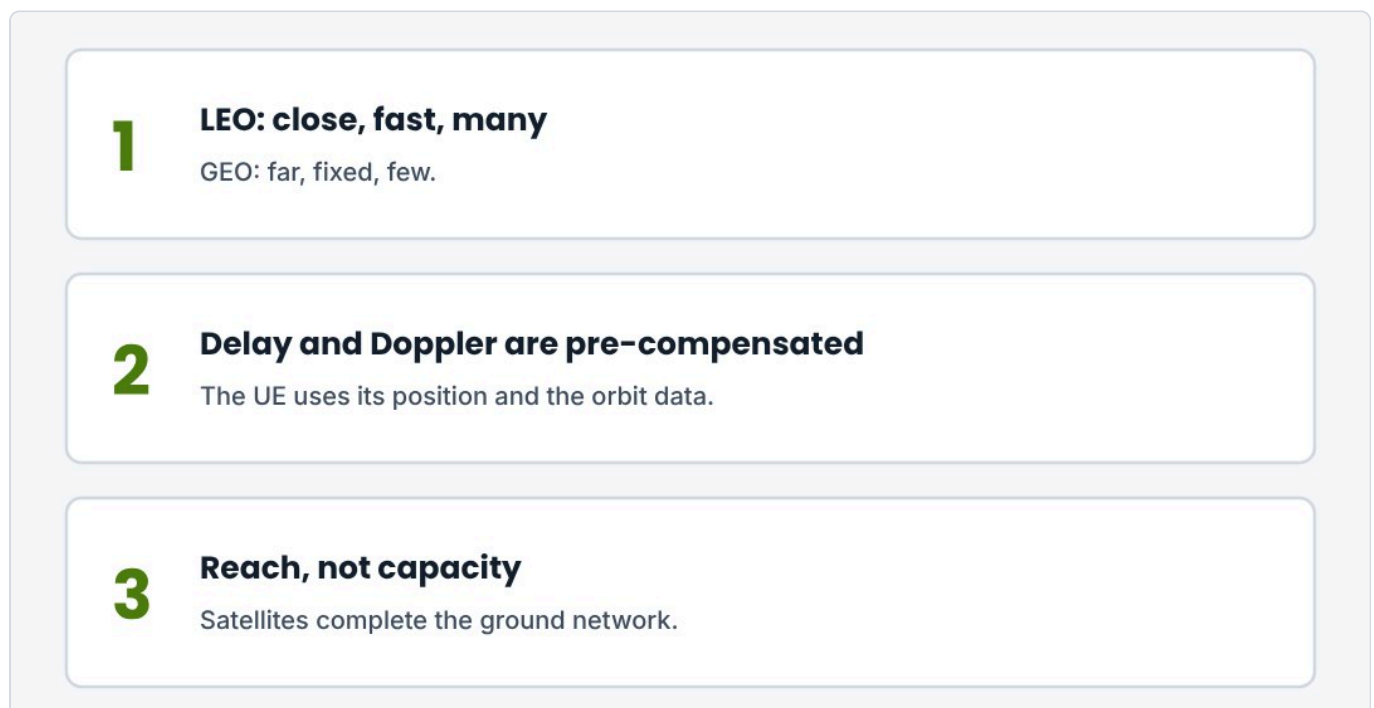
8.7 Use cases



Satellites will not replace ground networks: capacity per square kilometre is far lower. Their value is reach. They cover oceans, deserts and mountains, connect ships, aircraft and remote machines, and keep a minimum service alive when a disaster destroys the ground network.

3GPP TS 22.261 §6.46 · TR 22.822

8.8 Three things to remember



Orbit sets delay and speed. The payload can relay or host the base station. 6G aims at one design for ground and space.

3GPP TR 38.821 · Release 20 6G study

Ambient IoT, Energy and Sustainability

Devices without batteries, networks that sleep, and energy as a design requirement.

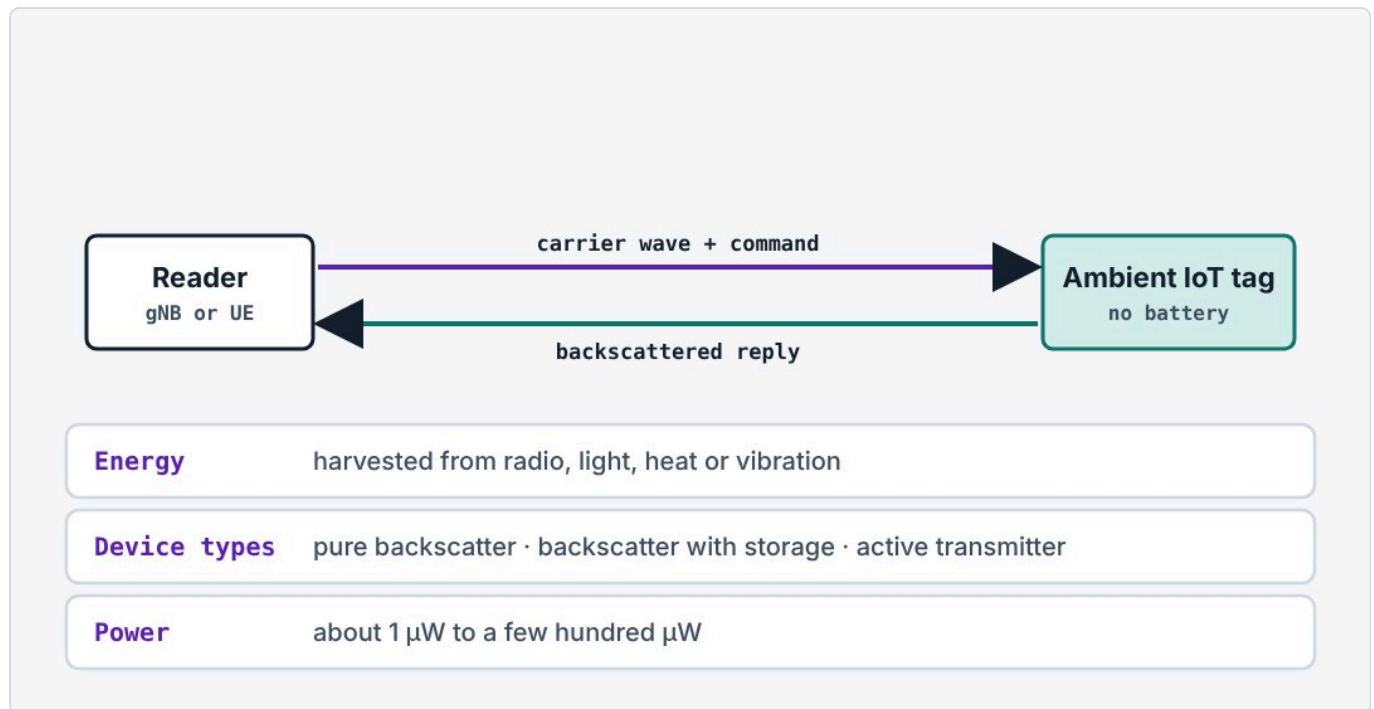
9.1 The range of IoT devices

CLASS	PEAK RATE	POWER SOURCE	TYPICAL USE
RedCap	tens of Mbit/s	battery, recharged	wearables, cameras
eRedCap	about 10 Mbit/s	battery	industrial sensors
LTE-M	about 1 Mbit/s	battery, years	trackers, meters
NB-IoT	tens of kbit/s	battery, 10 years	meters, sensors
Ambient IoT	below 10 kbit/s	harvested energy	tags, labels

Mobile IoT now spans several classes. RedCap is a reduced 5G device for wearables and sensors. LTE-M and NB-IoT serve low-rate devices for years on one battery. Ambient IoT goes one step further: devices so simple that they need no battery at all. 6G is meant to support the whole range from its first release.

3GPP TR 38.875 (RedCap) · TS 36.300 (NB-IoT, LTE-M) · TR 38.848 (Ambient IoT)

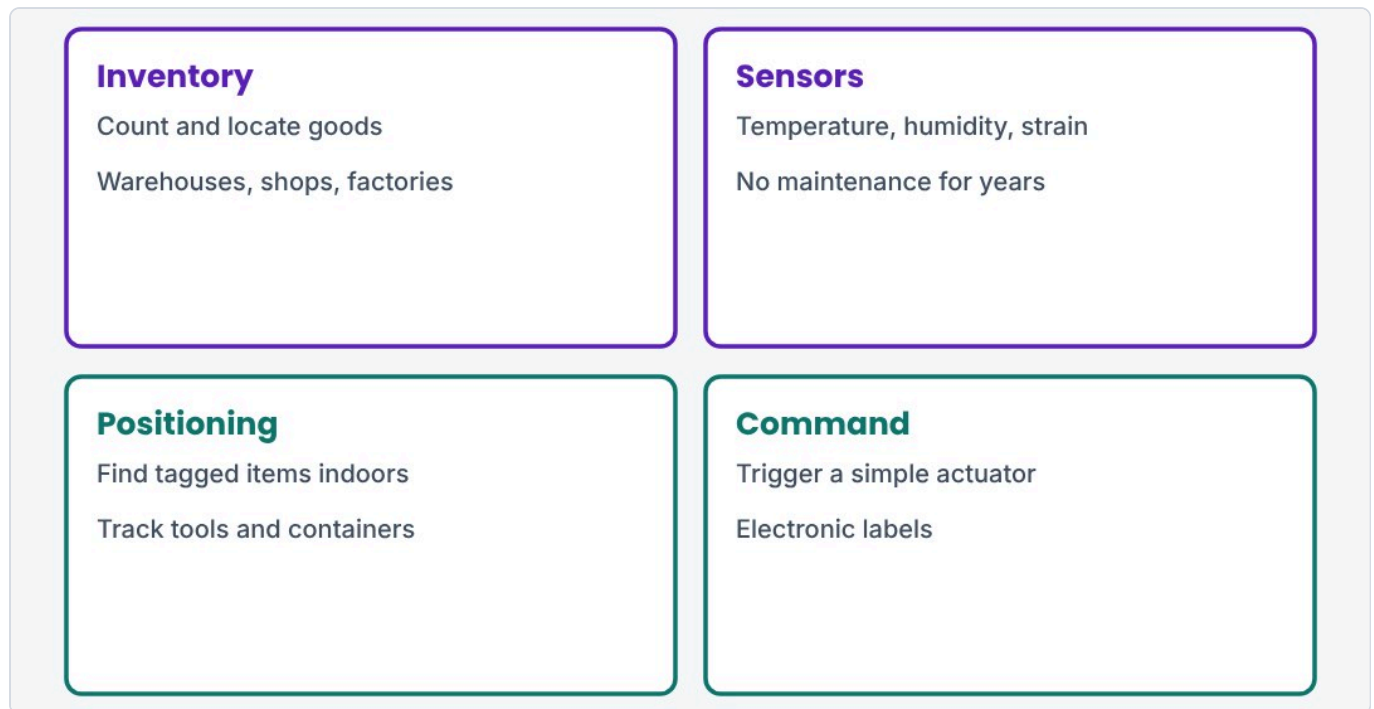
9.2 Ambient IoT: no battery



An ambient IoT device takes its energy from the surroundings: radio waves, light, heat or motion. Many do not generate a carrier of their own. They reflect a signal sent by the network and modulate the reflection, called backscatter. That keeps power consumption in the range of microwatts.

3GPP TR 38.848 · TR 38.769 (Release 19 study)

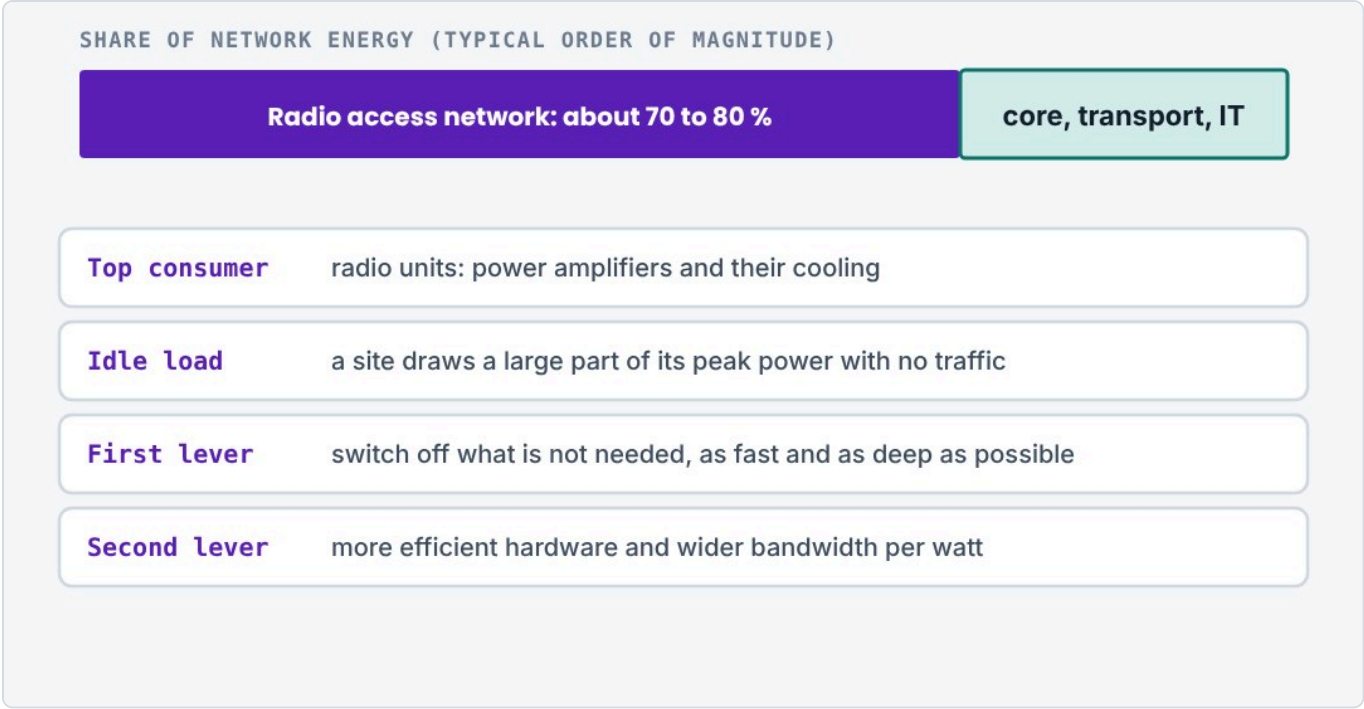
9.3 What ambient IoT is for



The target is objects that are too cheap, too many or too hard to reach for a battery device. Examples are pallets and parcels in logistics, stock in a warehouse, tools in a factory and sensors sealed inside structures. Release 19 specified a first indoor inventory use; 6G is to include it natively.

3GPP TS 22.369 · Release 19 Ambient IoT work item

9.4 Where the energy goes in a network



Most of a mobile network’s electricity is used in the radio access network, and there mostly in the power amplifiers and the always-on signals. A base station consumes a large share of its peak power even when it carries no traffic. So the first lever is to do less when idle.

3GPP TR 38.864 · operator and vendor energy reports

9.5 Energy-saving techniques

DOMAIN	TECHNIQUE
Time	longer periods for broadcast signals, cell DTX and DRX, micro-sleep
Frequency	carrier and cell switch-off, on-demand secondary cells
Space	switch off antenna elements and transceivers at low load
Power	lower transmit power, adapted to the users present
Device	wake-up signal, low-power wake-up receiver
System	AI-based load prediction to plan the sleep

Savings come in four domains. In time, the cell transmits less often. In frequency, carriers are switched off when load is low. In space, antenna elements are muted. In power, the output is reduced. 5G-Advanced introduced several of these, and 6G builds them into the basic design.

3GPP TR 38.864 · Release 18 and 19 network energy saving

9.6 Energy efficiency as a requirement

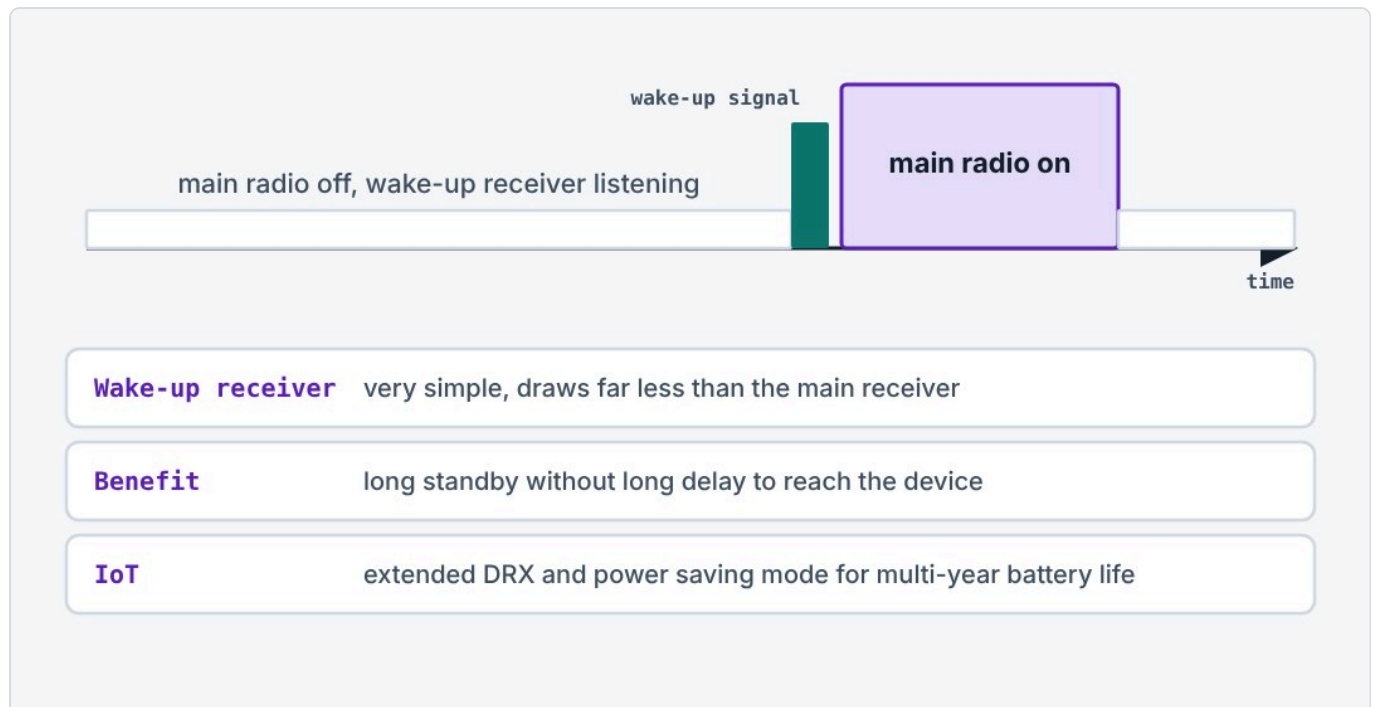
SUSTAINABILITY IN NUMBERS

Energy efficiency	bits delivered per joule, at a given load
Energy at low load	consumption with little or no traffic
Measured	by the network: energy as a KPI per function and slice
Exposed	energy information available to operators and customers
Life cycle	manufacturing, operation and recycling of equipment

IMT-2030 lists sustainability as a capability and as an overarching principle. For the network that means two things: energy per bit must fall as traffic grows, and absolute consumption at low load must fall too. 3GPP studies energy as a metric that the network measures, reports and can use as a service criterion.

ITU-R M.2160 §5 · 3GPP TS 28.554 (energy KPIs) · TR 22.882

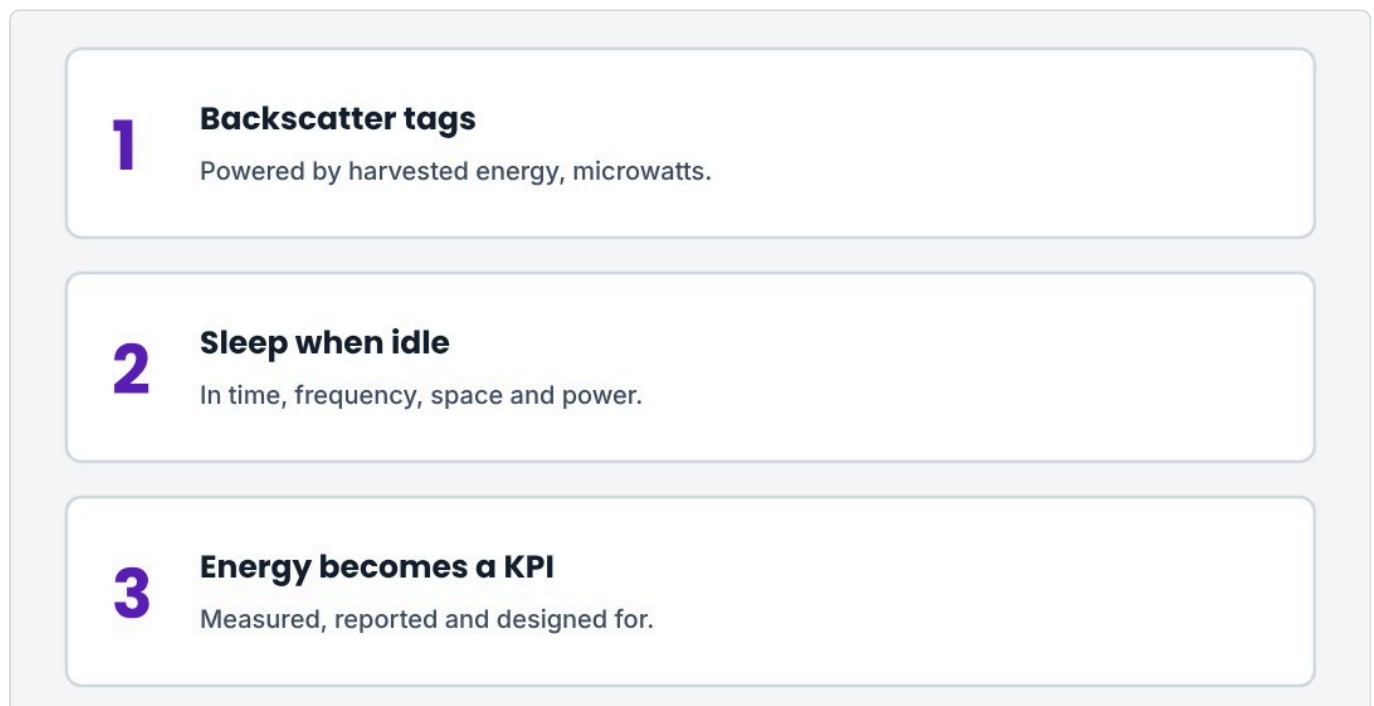
9.7 Device energy



Battery life matters as much as network energy. A device spends most of its time waiting. A low-power wake-up receiver lets the main radio sleep completely until the network sends a simple wake-up signal. For IoT, long sleep cycles and small signalling loads decide whether a battery lasts ten years.

3GPP TR 38.869 (low-power wake-up signal) · Release 19 LP-WUS

9.8 Three things to remember



IoT in 6G reaches down to battery-free tags. Network energy is mostly a radio and an idle-time problem. Sustainability is now a requirement, not a side effect.

3GPP TR 38.848 · TR 38.864 · ITU-R M.2160

Security, Privacy and Resilience

What 6G inherits from 5G security, what changes with quantum computers, AI and sensing, and what resilience means.

10.1 The 5G security baseline

CARRIED OVER FROM 5G

SUCI

the permanent identity is encrypted before it is sent

5G-AKA, EAP-AKA'

mutual authentication with home network control

User plane integrity

optional protection against modified data

Service-based security

TLS and OAuth 2.0 between core functions

SEPP

protected signalling between operators

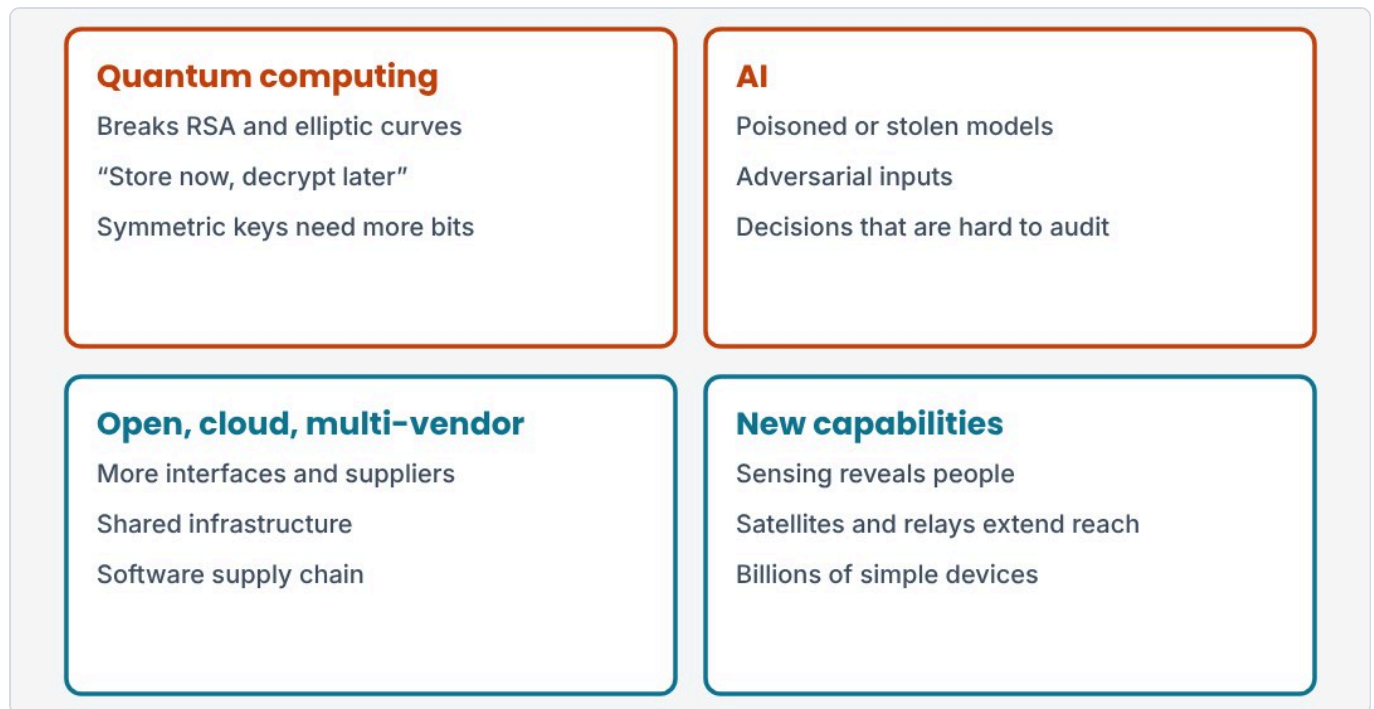
Slice authentication

extra authentication for a network slice

6G does not start from zero. 5G already hides the subscriber identity on the air, authenticates in both directions, can protect the integrity of user data and secures the interfaces between operators. The 6G work keeps these and looks at what new capabilities and new threats require.

3GPP TS 33.501

10.2 New threats



Three developments change the picture. Quantum computers could one day break the public-key methods used today. AI models bring their own weaknesses, such as poisoned training data. And a network that senses, runs in open clouds and includes satellites has a larger attack surface.

3GPP SA3 6G security study (Release 20) · ITU-R M.2160

10.3 Post-quantum cryptography

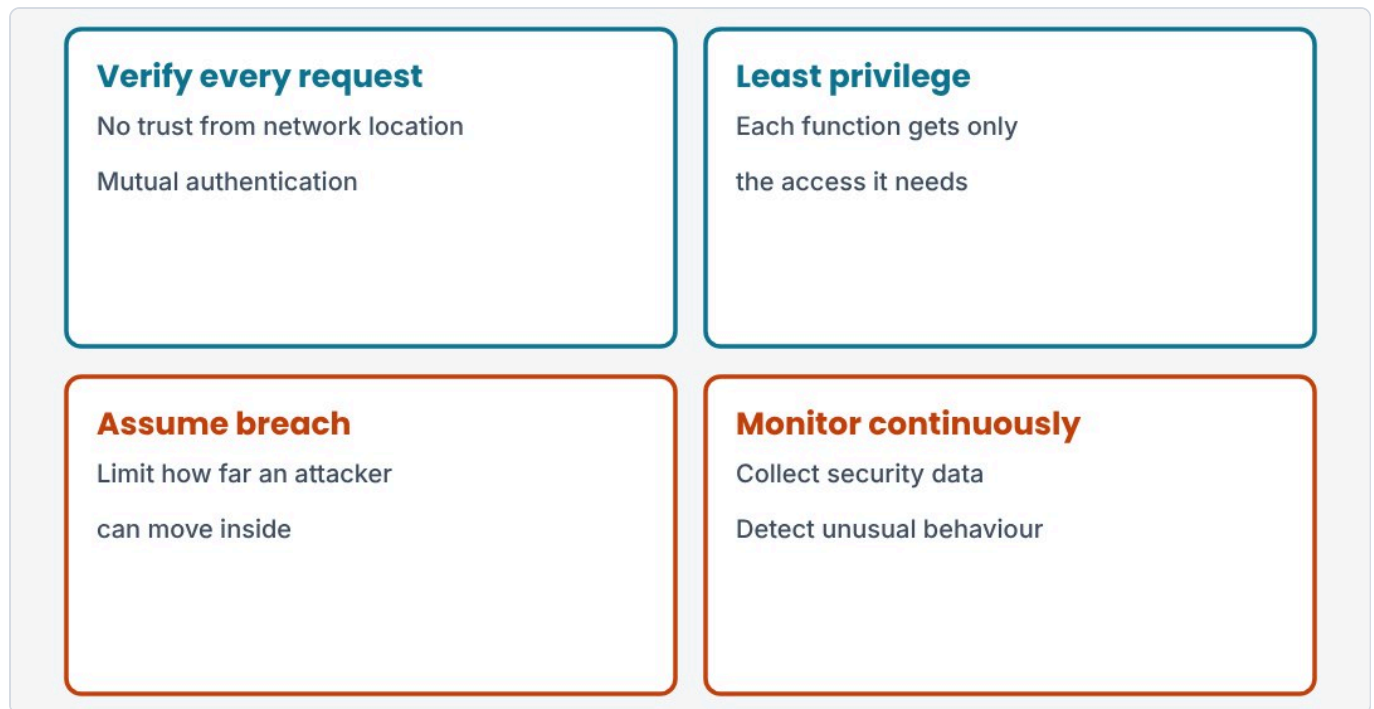
WHERE	TODAY	DIRECTION
SUCI concealment	elliptic curve (ECIES)	post-quantum KEM
TLS, IPsec key exchange	elliptic curve DH	hybrid: classic + ML-KEM
Certificates	RSA, ECDSA	ML-DSA and others
Radio ciphering	128-bit AES, SNOW, ZUC	256-bit variants
AKA	symmetric, 128-bit	256-bit keys

ML-KEM = FIPS 203 · ML-DSA = FIPS 204

The answer to the quantum threat is new public-key algorithms that quantum computers are not expected to break. NIST published the first standards in 2024. Mobile networks use public-key methods to conceal the identity and in TLS and IPsec. Symmetric algorithms stay but move to 256-bit keys.

NIST FIPS 203, 204, 205 (2024) · 3GPP TR 33.841 · TS 33.501 Annex C

10.4 Zero trust



Earlier generations trusted everything inside the operator network. With cloud platforms and many suppliers that assumption no longer holds. Zero trust means that every request between functions is authenticated and authorised, access is limited to what is needed, and behaviour is monitored continuously.

NIST SP 800-207 · 3GPP TR 33.894

10.5 Privacy in a sensing, AI-driven network

CAPABILITY	PRIVACY QUESTION
Sensing	who may sense what, and how are bystanders protected?
Positioning	who may request centimetre-level location?
AI and data	which data is collected, and can it be linked to a person?
Exposure APIs	what does a third party learn about users?
Digital twins	how detailed may a model of people's movement be?

6G raises privacy questions that go beyond subscribers. Sensing can detect people who carry no device and never agreed to anything. AI needs data that may reveal behaviour. Very precise positioning shows where a person is to within centimetres. Rules for consent, purpose and access have to be part of the design.

3GPP TS 22.137 §5 (sensing privacy) · TR 22.870 · GDPR principles

10.6 Resilience



Resilience means that the network keeps delivering an acceptable service when something goes wrong, and recovers quickly. Causes can be a software fault, a power cut, a natural disaster or an attack. IMT-2030 names it as a capability. Satellites, spare capacity and the ability to isolate a faulty part all contribute.

ITU-R M.2160 §5 · 3GPP TR 22.870

10.7 Trust in AI decisions

Data integrity	protect training data against poisoning
Model integrity	signed models, verified before deployment
Robustness	tested against adversarial inputs
Monitoring	detect drift and abnormal outputs
Fallback	a non-AI procedure is always available
Accountability	logs that explain which model decided what

When AI controls parts of the network, security includes the models themselves. Training data must be protected against manipulation. Models must be checked before use and watched while they run. And for important decisions a human or a rule-based function must be able to overrule the model.

3GPP TR 33.898 · EU AI Act principles · NIST AI RMF

10.8 Three things to remember

- 1 Build on 5G**
SUCI, mutual authentication, protected interfaces.
- 2 Prepare for quantum computers**
ML-KEM, ML-DSA and 256-bit symmetric keys.
- 3 Protect people who have no device**
Sensing needs its own privacy rules.

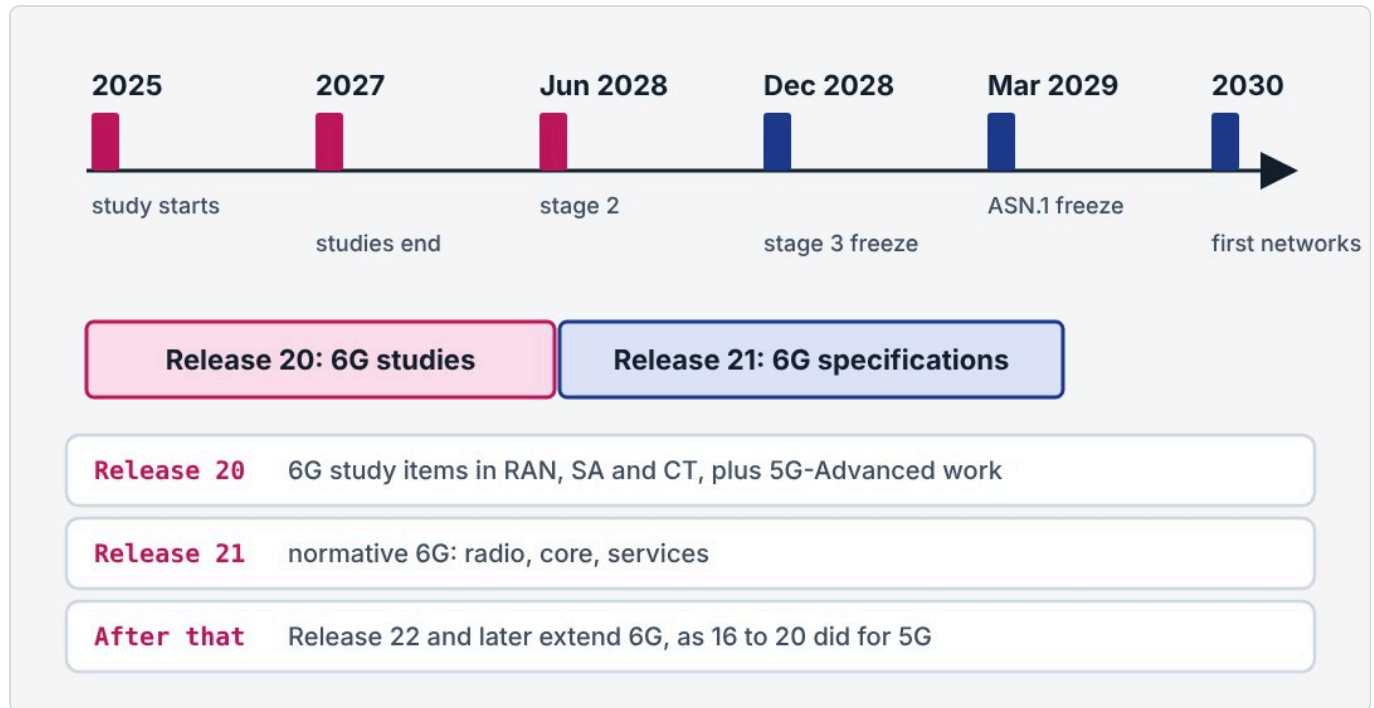
The 5G security architecture continues. Cryptography moves to post-quantum and 256-bit. Privacy for sensing and trust in AI are new topics.

3GPP TS 33.501 · SA3 6G study

Roadmap, and What to Prepare For

The 3GPP and ITU schedule up to the first networks, and the skills a test engineer will need.

11.1 The 3GPP schedule



3GPP works in two steps. Release 20 studies 6G and, in parallel, completes 5G-Advanced. Release 21 writes the first 6G specifications. The timeline approved in June 2026 puts the final stage 3 freeze in December 2028 and the last code freeze in March 2029.

3GPP TSG plenaries June 2026: Release 21 timeline (public reports)

11.2 How 3GPP and the ITU fit together

YEAR	ITU-R	3GPP
2023	framework M.2160	workshops, Release 20 planning
2025 – 26	requirements, evaluation method	Release 20 studies
2027	submissions open, WRC-27	studies end, Release 21 starts
2028	submissions continue	Release 21 specifications
2029	submission deadline, evaluation	final freeze, self-evaluation
2030	IMT-2030 specifications	Release 22, first networks

The two schedules interlock. The ITU opens the window for candidate technologies in 2027. 3GPP submits a description of its 6G radio, first as a preliminary version and then completed with the Release 21 specifications. Evaluation follows, and the ITU approves the IMT-2030 specifications in 2030.

ITU-R WP 5D IMT-2030 process · 3GPP planning

11.3 Decided, emerging, open

Agreed in the study

CP-OFDM, 5G channel codes

3 to 400 MHz carriers

Standalone baseline

CU / DU split possible

Direction clear

Spectrum sharing with 5G

Native NTN and IoT

AI life cycle framework

Energy-saving design

Open

WRC-27 bands and conditions

Sensing procedures

Core changes

Device categories

Working agreements only: nothing is normative before Release 21

It helps to sort what is known. Some radio basics are agreed in the study. Other topics have a clear direction but no detail. And for several the outcome is open, especially spectrum and the details of sensing and AI. Anything can still change until Release 21 is frozen.

Status as of October 2026

11.4 From 5G-Advanced to 6G

IN 5G-ADVANCED	CONTINUES IN 6G AS
AI/ML for beam, CSI, positioning	AI-native air interface
Network energy saving	lean carrier design
NR over satellite (NTN)	unified terrestrial and satellite access
RedCap, Ambient IoT	all IoT classes from the first release
Subband full duplex	duplex evolution
ISAC channel model	integrated sensing and communication
Sidelink, positioning	native positioning and device-to-device

Many 6G features do not appear suddenly. They start in 5G-Advanced and continue. An engineer who works with Releases 18 and 19 today is already learning the first version of what 6G will make native.

3GPP Releases 18, 19, 20

11.5 What changes for testing

AREA	NEW TEST QUESTION
Upper mid band	over-the-air tests with arrays of a thousand elements
AI features	performance over data sets, not one expected output
Sensing	emulated targets with defined range, speed and reflection
NTN	emulated delay, Doppler and moving cells
Energy	measuring consumption as a pass or fail criterion
Spectrum sharing	5G and 6G on one carrier: coexistence tests

New capabilities need new test methods. Radio tests at 7 to 15 GHz with large arrays are done over the air, without cables. AI features cannot be verified against a fixed expected output. Sensing needs targets and scenes that can be reproduced. Satellite links need emulation of delay and Doppler.

3GPP RAN4 and RAN5 work: OTA testing, AI/ML testing, NTN

11.6 Skills worth building now

5G standalone in depth

NR physical layer and procedures

5G core, slicing, QoS

Machine learning basics

Training, inference, validation

Model monitoring and drift

Radar and positioning

Range, Doppler, angle

OFDM radar processing

Cloud and automation

Containers, CI/CD, observability

Scripting and data analysis

You do not have to wait for the specifications. The foundations of 6G are known. Solid knowledge of the 5G standalone system comes first, because 6G evolves from it. On top come the basics of machine learning, radar signal processing, satellite links and cloud platforms.

Summary of the previous chapters

11.7 How to follow the work

3GPP work plan	which study and work items exist, and their status
TSG plenaries	every quarter: decisions on scope and timeline
Technical reports	study results: 38-series radio, 22- and 23-series system
ITU-R WP 5D	IMT-2030 requirements, evaluation, submissions
WRC-27	October to November 2027: the spectrum outcome
O-RAN ALLIANCE	open interface and lower-layer split specifications

The process is public. 3GPP publishes meeting reports, technical reports and the work plan. The ITU publishes the IMT-2030 documents. Reading the summaries after each plenary meeting is the most efficient way to stay current, and the Release 20 technical reports will hold the detailed study results.

3gpp.org · itu.int IMT-2030 pages

11.8 Three things to remember

- 1 December 2028**
Planned stage 3 freeze of the first 6G release.
- 2 5G-Advanced is the on-ramp**
Releases 18 to 20 introduce the building blocks.
- 3 This guide is a snapshot**
Status of October 2026. Check current sources.

Specifications at the end of 2028, networks around 2030. Much of 6G starts in 5G-Advanced. The best preparation is deep 5G knowledge plus AI, sensing and cloud basics.

[3GPP Release 21 timeline](#) · [ITU-R IMT-2030 process](#)